Chi's Conjecture for Split Reductive Groups

A Complete Proof Argument under He's Assumptions

English research-manuscript version

Abstract

This manuscript assembles the written proof argument for the top-dimensional component formula at all Newton points of a split connected reductive group in the equal- and mixed-characteristic settings of He. It expands the general linear and symplectic lattice arguments, the ordinary and vector orthogonal cases, the spinorial finite lemmas, the exceptional finite lemmas, integral conjugacy and tame descent, and transfer through all reductive central forms. The finite-dimensional structural theorems and He's geometric reduction are stated as explicit inputs.

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All references to He's theorem and proposition numbers use *Irreducible components of affine Lusztig varieties*, arXiv:2502.16441v1 [He].

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1 The Statement, Notation, and Standard Inputs

1.1 The ground field and the group

Let k be algebraically closed, and take

$$(\mathcal{O}, L, t) = \begin{cases} (k[[t]], k((t)), t), & \text{in equal characteristic,} \\ (W(k), W(k)[1/p], p), & \text{in He's mixed-characteristic setting.} \end{cases}$$

Normalize the valuation by $v_L(t) = 1$. Let G be a split connected reductive group over L, and assume

$$\operatorname{char} k = 0 \quad \text{or} \quad p = \operatorname{char} k > 0, \qquad p \nmid |W(G)|.$$

The Weyl group of every Levi is a subgroup of $W(G)$; hence the same characteristic hypothesis holds in all Newton Levis. The orders of the finite centers, tame extensions, and finite actions used below are invertible in $\mathcal{O}$.

Fix a split maximal torus T_0, a Borel, and a dominant Weyl chamber. Write

$$X_* = X_*(T_0), \qquad Q^\vee = \mathbb{Z}\Phi^\vee, \qquad \pi_1(G) = X_*/Q^\vee.$$

All torsion in $\pi_1(G)$ is retained. Let K_{sp} be the standard hyperspecial subgroup, let $I \subset K_{\mathrm{sp}}$ be an Iwahori subgroup, let $\widetilde{W} = X_* \rtimes W_0$, and let w_0 be the longest element of W_0. Denote the length-zero subgroup by Ω.

1.2 Affine Lusztig varieties and the component formula

For a regular semisimple $\gamma \in G(L)$ and a dominant coweight μ, put

$$Y_\mu(\gamma) = \{gK_{\mathrm{sp}} \in G(L)/K_{\mathrm{sp}} : g^{-1}\gamma g \in K_{\mathrm{sp}} t^\mu K_{\mathrm{sp}}\}.$$

In the affine flag variety put

$$Y_w(\gamma) = \{gI : g^{-1}\gamma g \in I\dot{w}I\}.$$

In mixed characteristic these expressions have the perfected geometric interpretation used by He. Write $\Sigma^{\mathrm{top}} X$ for the irreducible components of dimension $\dim X$.

Associate $\{\gamma\}$ with the corresponding σ-conjugacy class $[b]$ as in He, Section 1.2; their Newton and Kottwitz invariants match. Let λ_b be the best integral approximation used in He, Section 3.2. Let $\widehat{G}$ be the complex Langlands dual group, and let V_μ be its irreducible highest-weight representation of highest weight μ.

Theorem 1.1 (The claimed main formula)**.** *Under the assumptions above, for every regular semisimple γ and every dominant μ with $Y_\mu(\gamma) \neq \varnothing$, the assembled argument gives*

$$\boxed{\#\left(Z_G(\gamma)(L)\backslash \Sigma^{\mathrm{top}} Y_\mu(\gamma)\right) = \dim V_\mu(\lambda_b).}$$

No integral-Newton condition is imposed. Arbitrary split central tori, central quotients, and central couplings between simple factors are allowed.

1.3 The best integral approximation retains Kottwitz data

In the split case one can describe λ_b concretely. Choose $\lambda_0 \in X_*$ mapping to $\kappa(b)$, and write

$$\nu_b - \lambda_0 = \sum_i c_i \alpha_i^\vee, \qquad c_i \in \mathbb{Q},$$

in the semisimple directions; the central projection of ν_b equals the rational image of $\kappa(b)$. Then

$$\lambda_b = \lambda_0 + \sum_i \lfloor c_i \rfloor \alpha_i^\vee.$$

Changing λ_0 changes the c_i by integers, so the expression is independent of that choice. It is the largest integral element in the Kottwitz coset lying below ν_b in the coroot order. Equivalently,

$$\nu_b - \lambda_b = \sum_i d_i \alpha_i^\vee, \qquad 0 \le d_i < 1.$$

Thus λ_b is integral but generally differs from the rational Newton vector. Torsion Kottwitz data cannot be omitted. The weight space $V_\mu(\lambda_b)$ uses this integral weight.

1.4 The required local property

For a basic regular semisimple $\eta \in M(L)$ in a split reductive group M, let τ be the length-zero element of its Kottwitz class, and let $P = P_K$ be a τ-stable connected parahoric. Put

$$\mathcal{F}^{\mathrm{reg}}_{P,\tau}(\eta) = \{gP : g^{-1}\eta g \in P\dot{\tau}, \text{ with regular reductive-coset image}\}.$$

Regularity means minimum centralizer dimension in the relevant component of the disconnected reductive quotient. The parahoric is *very special* when its log-volume

$$n_K = \ell(w_K)$$

is maximal among all τ-stable spherical subsets. Basic means that the Newton vector is central in M, not that the Kottwitz class is zero.

Write $\mathsf{L}(M)$ for the local assertion

$$T_\eta(L) \text{ is transitive on } \mathcal{F}^{\mathrm{reg}}_{P,\tau}(\eta)(k), \qquad T_\eta = Z_M(\eta)^\circ.$$

For regular semisimple η, T_η is a maximal L-torus. The identity-component condition is necessary in upward central-isogeny transfer.

1.5 The standard theorems used as inputs

The following are the explicit standard inputs of the manuscript.

A. He's geometric reduction. Use Theorems 1.3, 1.4, 2.8, and 3.2, Propositions 2.3, 2.5, and 2.7, and the component correspondences in Section 3.6. For the ordinary hyperspecial case use Theorem 3.5. Section 2 explains these uses; the integral-Newton conclusion is not applied in advance to nonintegral classes.

B. The finite-dimensional Kostant regular quotient. For the split orthogonal Lie algebras used here, the quotient is smooth on the regular locus, its geometric regular fibers are single special-orthogonal orbits, and regular centralizers are smooth. In even dimension the quotient includes the Pfaffian, not just the characteristic polynomial. Section 5 derives the integral consequence.

C. The finite-dimensional Steinberg regular quotient. For a split simply connected semisimple group in the permitted characteristic, the regular quotient is smooth, geometric regular fibers are single conjugacy orbits, and full regular centralizers are smooth. Semisimple centralizers are connected. The integral transporter consequence is proved in Section 8.

D. Tame finite-order automorphisms. An automorphism of a connected reductive group over an algebraically closed field, of finite order invertible in that field, stabilizes a Borel and a maximal torus therein. Its identity fixed group is reductive. Tame H^1 of a unipotent group is trivial; vector-group cohomology is killed by averaging. The finite proofs distinguish this stable-Borel input from the unique Borel available after regular-unipotent regularity has been proved.

E. Tame fixed group models. An invertible-order finite action on a smooth group scheme has a smooth fixed model. Root valuation bounds identify the connected parahoric; the fixed jet kernel is connected unipotent. Section 8 records the root parameters, reductive quotients, and full normalizer grades.

F. The full torus Kottwitz sequence. Over the complete strictly henselian field under consideration, the relevant torus has a surjective map

$$T(L) \longrightarrow X_*(T)_I.$$

The coinvariants include torsion. Its kernel is the connected Neron/parahoric subgroup, which need not be the whole bounded subgroup. Section 3 proves the consequence for all group Kottwitz grades.

We also use Hensel's lemma, Hilbert 90, split root-group structure, the classical regular-unipotent Jordan forms, and the finite Weyl classifications stated in Section 9. These are finite-dimensional or local-algebraic inputs, not replacement assumptions for the desired component formula.

2 Reduction to Local Regular-Point Transitivity

2.1 The actual top-contributing reduction paths

Set $w = w_0 t^\mu$, choose a reduction tree $\mathcal{T}$, and let $\mathcal{R}$ be the paths matching $[b]$ which actually contribute to top dimension:

$$\ell(\mathfrak{p}) + \ell(\mathrm{end}(\mathfrak{p})) - \langle \nu_b, 2\rho \rangle = \dim X_w(b).$$

The associated invariants in He, Section 1.2, make the endpoint matching conditions identical on both sides. Theorem 1.4 supplies the same dimension shift for every nonempty stratum:

$$\dim Y_w(\gamma) - \dim X_w(b) = \dim Y_\gamma.$$

Consequently the actual top-path sets agree. Theorem 2.8 and Proposition 2.5 give

$$\#\big(Z_G(\gamma)(L)\backslash \Sigma^{\mathrm{top}} Y_w(\gamma)\big) = \sum_{\mathfrak{p}\in\mathcal{R}} n_{\mathfrak{p},\gamma},$$

$$\#\big(J_b\backslash \Sigma^{\mathrm{top}} X_w(b)\big) = |\mathcal{R}|.$$

No virtual-dimension equality and no regularity of μ are assumed.

2.2 Every actual top endpoint is very special

For $\mathfrak{p} \in \mathcal{R}$, the stratum $X_{\mathfrak{p}}(b)$ is J_b-stable and locally closed, with

$$\dim X_{\mathfrak{p}}(b) = \dim X_w(b).$$

The closure of a top-dimensional irreducible component of the stratum is a top component of the ambient variety. Such a locally closed piece is dense open in its closure, so the stabilizers agree. Closure therefore gives a J_b-equivariant component injection preserving stabilizers.

He, Proposition 2.3, identifies the stratum components equivariantly with the endpoint components. The flag–Grassmannian correspondence quoted in Section 3.2 identifies $\Sigma^{\mathrm{top}} X_{w_0 t^\mu}(b)$ with $\Sigma^{\mathrm{top}} X_\mu(b)$. By Theorem 3.2(2), their stabilizers are very special parahorics of J_b. The endpoint stabilizer is the corresponding $P_{\mathfrak{p}}(\breve{F}') \cap J_b$, as in that section. Hence the endpoint parahoric is very special in the combinatorial log-volume sense.

Only actual top dimension is used. In He's cordial-path argument, cordiality supplies this dimension equality; it is not inserted here as an additional virtual-dimension hypothesis.

2.3 Passing to the actual Newton Levi

Write the endpoint as $u\tau$, with u τ-elliptic in W_K. Theorem 1.3 moves the fixed element to a basic $\eta \in M(L)$ in $P_K \dot{\tau}$, with the required endpoint-fiber and full-centralizer correspondences. Here M is the actual split Newton Levi, including any central coupling.

Proposition 2.7 and its proof show that the nonregular complement has strictly smaller dimension and identify the top components with those of the regular partial fiber. Therefore $\mathsf{L}(M)$ implies

$$n_{\mathfrak{p},\gamma} = 1.$$

To justify the passage from points to components, choose in each top component a k-point lying in no other component. Such open parts have k-points in the locally finite-type spaces involved, and the same argument applies to their mixed-characteristic perfections. A point transporter carries the unique component through the first point to the unique component through the second. Thus the identity centralizer, and hence the full centralizer, has one component orbit.

2.4 Returning to the Grassmannian

He, Section 3.6, uses the fibration from the flag strata indexed by $W_0 t^\mu W_0$ to $Y_\mu(\gamma)$, with fiber K_{sp}/I. Its general dimension comparison shows that all strata except $w_0 t^\mu$ have smaller dimension; this comparison does not use integral Newton. It yields the equivariant top-component correspondence.

Combining it with the Chen–Zhu formula in Theorem 3.2(1) gives

$$C_\mu(\gamma) := \#(Z_G(\gamma)(L) \backslash \Sigma^{\mathrm{top}} Y_\mu(\gamma)) = \sum_{\mathfrak{p} \in \mathcal{R}} n_{\mathfrak{p},\gamma}, \qquad |\mathcal{R}| = \dim V_\mu(\lambda_b).$$

It remains to establish L at every very special basic endpoint of every actual Newton Levi.

3 Centralizer Grades and All Reductive Central Forms

3.1 Every Kottwitz grade occurs in the identity centralizer

Lemma 3.1. *For a split reductive H and regular semisimple $\eta \in H(L)$, put $T_\eta = Z_H(\eta)^\circ$. Then*

$$\kappa_H(T_\eta(L)) = \pi_1(H).$$

Proof. The full torus Kottwitz sequence gives $T_\eta(L) \twoheadrightarrow X_*(T_\eta)_I$. Over a separable closure, identify T_η with a split maximal torus. Its lattice surjects onto the quotient by coroots $\pi_1(H)$. Inertia acts through $W(H)$, and Weyl reflections act trivially on that quotient. Hence the map factors through a surjection

$$X_*(T_\eta)_I \twoheadrightarrow \pi_1(H).$$

Kottwitz functoriality proves the assertion. □

Torsion cannot be discarded: a ramified quadratic norm-one torus can have coinvariants $\mathbb{Z}/2$, although all its points are bounded. Its connected Neron subgroup still has index two. The characteristic assumption also ensures that the inertia actions here are tame.

3.2 The image of a finite central isogeny

Let $f : H' \to H$ be a finite central isogeny of split connected reductive groups with kernel order invertible in k, and set

$$B = f_*\pi_1(H') \subset \pi_1(H).$$

The groups share the coroot lattice, so f_* is injective. For corresponding connected parahorics P', P,

$$f(H'(L)) = \kappa_H^{-1}(B), \qquad f(P') = P, \qquad f^{-1}(P) = P'.$$

Root groups lift isomorphically, Weyl representatives lift through root groups, and split-torus units lift by invertible-degree Hensel root extraction. The remaining valuation obstruction is exactly the cocharacter class modulo the shared coroot lattice. Root bounds and unit torus models coincide at corresponding parahorics. The facet-stabilizer description intersected with the Kottwitz kernel, together with injectivity on π_1, gives the inverse-image equality. Separable central isogenies preserve residue-coset regularity.

3.3 Downward transfer: the fixed element must lift

Suppose $\eta \in H(L)$ has $\kappa_H(\eta) \in B$, and choose its lift $\eta' \in H'(L)$ and compatible length-zero representatives. The preimage of the fixed target coset is the fixed source coset. For two target points g_iP, Lemma 3.1 gives $z_i \in T_\eta(L)$ of grade $\kappa_H(g_i)$. The adjusted representatives $z_i^{-1}g_i$ have grade zero and lift to $h_i \in H'(L)$ in the regular fiber for the fixed η'.

If the source identity torus is transitive, a source transporter $c' \in Z_{H'}(\eta')^\circ(L)$ produces

$$z_2 f(c') z_1^{-1} \in T_\eta(L)$$

carrying the original target points. The fixed-element condition is indispensable. Grade translation moves variable flags; it cannot change the Kottwitz class of the fixed element.

3.4 Upward transfer for an already given source element

Conversely, suppose target identity-torus transitivity is known and $\eta' \in H'(L)$ is already given. A target identity-torus transporter between the images of two source flags has grade the difference of two source-allowed grades, hence lies in B and lifts. The full inverse image of the target identity torus is the source maximal torus containing η'; its lift therefore centralizes η'. The equality $f^{-1}(P) = P'$ retains the source flag coset. No connectedness of the full target centralizer is assumed.

3.5 A split central-torus quotient

Suppose

$$1 \longrightarrow E \longrightarrow H \xrightarrow{q} M \longrightarrow 1$$

is central with E split. Hilbert 90 lifts every fixed element and variable flag, and corresponding connected parahorics map onto one another. Root data give

$$0 \longrightarrow X_*(E) \longrightarrow \pi_1(H) \longrightarrow \pi_1(M) \longrightarrow 0.$$

Injectivity follows because central cocharacters have zero intersection with the rational coroot span.

Lift the fixed element to η_H, and choose τ_H with its full grade. A lift g_H of a target fiber point satisfies

$$g_H^{-1}\eta_H g_H \in P_H \dot{\tau}_H E(L).$$

The grades on the left and of τ_H agree, so the extra E-factor has zero valuation and lies in $E(\mathcal{O}) \subset P_H$. Thus the lift belongs to the correct fixed fiber. Central-kernel dimensions add equally to actual and minimum coset-centralizer dimensions, so regularity is equivalent. Source identity-torus transitivity descends to M.

3.6 A construction retaining arbitrary central coupling

Let M be any split connected reductive group. Let S be the simply connected cover of its derived group and let $T_c = Z(M)^\circ$ be its actual embedded connected center. Write

$$M = (S \times T_c)/C.$$

The finite C projects injectively to $Z(S)$, since a kernel element $(1, t_c)$ has $t_c = 1$. Embed its projection into a split torus E, by j, and form

$$H = (S \times T_c \times E)/\{(s_c, t_c, j(s_c)) : (s_c, t_c) \in C\}.$$

Then $H \to M$ has split central kernel E, and $H_{\mathrm{der}} = S$, since the displayed finite subgroup intersects $S \times 1 \times 1$ trivially. The whole center of H need not be connected.

Put $A_H = H/H_{\mathrm{der}}$. The natural map

$$f : H \longrightarrow D = H_{\mathrm{ad}} \times A_H$$

is a finite central isogeny with kernel exactly $Z(S)$. Since H_{der} is simply connected,

$$\pi_1(H) \xrightarrow{\sim} X_*(A_H).$$

Consequently its image in $\pi_1(H_{\mathrm{ad}}) \times X_*(A_H)$ is a graph over the second coordinate, retaining all finite central coupling.

If every simple adjoint factor satisfies the local assertion, so does its product with the split torus A_H; the latter translates its flag cocharacter lattice. Upward finite-isogeny transfer gives the assertion in H, and split-torus descent gives it in M. The construction applies afresh to each actual Newton Levi.

The primes dividing $|Z(S)|$ divide the derived Weyl-group order and hence the ambient one, so no additional inseparable isogeny is introduced. Pure torus presentations impose no extra characteristic-degree isogeny requirement. It suffices to prove the local property for every simple adjoint type, every basic Kottwitz class, and every very special endpoint.

4 Types A and C

This section proves the local assertions needed for types A and C. Write $O = O_L$, choose a uniformizer t, and normalize the valuation by $v_L(t) = 1$. For a very special τ-stable parahoric subgroup P_K, write

$$\mathscr{Y}^{\mathrm{reg}}_{K,\tau}(\gamma) = \{gP_K : g^{-1}\gamma g \in P_K\dot{\tau},\ \overline{g^{-1}\gamma g} \text{ is regular in the residual coset}\}.$$

Here *very special* means that the length of the longest element of the corresponding finite Weyl group is maximal among all τ-stable spherical subsets. The conclusion of this section is that, for every relevant Newton Levi subgroup, the identity-component centralizer torus $T(L) = Z_G(\gamma)^\circ(L)$ acts transitively on this regular locus. The component-counting reduction of Section 2 then shows that every top-dimensional reduction path contributes 1.

All integer denominators, square roots, and finite-dimensional Jordan-form calculations for classical groups are taken under the characteristic assumptions of this manuscript. In particular, for GL_N, all primes at most N are excluded; for Sp_{2n} or GSp_{2n}, the prime 2 and all primes at most n are excluded.

4.1 All very special endpoints of a basic general linear block

Proposition 4.1. *Let $N = md$, and let $\gamma \in \mathrm{GL}_N(L)$ be regular semisimple. Suppose that all Newton slopes of its natural representation are equal to the reduced fraction r/d, where $d > 0$ and $\gcd(r, d) = 1$. Every very special endpoint is represented by an indexed periodic lattice chain*

$$\Lambda_{i+1} \subset \Lambda_i, \qquad \Lambda_{i+d} = t\Lambda_i, \qquad \dim_k(\Lambda_i/\Lambda_{i+1}) = m, \qquad i \in \mathbb{Z}.$$

The corresponding local coset condition is

$$\gamma\Lambda_i = \Lambda_{i+r} \qquad (i \in \mathbb{Z}).$$

Proof. For $N \geq 2$, label the vertices of the affine A_{N-1} diagram by $\mathbb{Z}/N\mathbb{Z}$. Let ω be the length-zero rotation whose determinant has valuation 1. The determinant valuation of the basic block is mr, so $\tau = \omega^{mr}$. Its vertex orbits are precisely the congruence classes modulo m, each containing d vertices.

Any τ-stable spherical subset must omit at least one such orbit. If q congruence classes are omitted, let $a_1, \ldots, a_q$ be the cyclic gaps between successive omitted vertices in a period of length m. Then $a_j \geq 1$ and $\sum_j a_j = m$. The remaining connected components contain d copies of A_{a_j-1} for each j. The length of the longest element is therefore

$$n_K = d\sum_{j=1}^{q} \frac{a_j(a_j-1)}{2}.$$

If $q > 1$, the identity

$$m(m-1) - \sum_j a_j(a_j-1) = 2\sum_{i<j} a_i a_j > 0$$

shows that this length is strictly less than $dm(m-1)/2$. Consequently, the very special subsets omit exactly one orbit. The corresponding partial complete lattice chain has d steps in each period, each of length m, and hence has the form asserted in the proposition. Choosing a different omitted orbit merely changes the reference chain and leaves these relations unchanged.

The length-zero element ω^{mr} shifts this chain by r partial chain indices, so its coset condition is exactly $\gamma\Lambda_i = \Lambda_{i+r}$. If $m = 1$, the remaining spherical subset is empty; if $d = 1$, the subgroup is the usual hyperspecial lattice stabilizer. The rootless case $N = 1$ has the same lattice description. □

4.2 A Hensel root in the fixed integral algebra

We next eliminate the slope denominator entirely within linear algebra, without changing the original lattice-point problem.

Lemma 4.2. *Under the assumptions of Proposition 4.1, one can construct commuting operators h, π from the fixed element γ such that*

$$\pi^d = t, \qquad \gamma = \pi^r h, \qquad h \in O[t^{-r}\gamma^d]^\times.$$

If $F = L[\pi]$, then F/L is a totally ramified extension of degree d, $O_F = O[\pi]$, and

$$L[\gamma] = F[h], \qquad Z_{\mathrm{GL}_N(L)}(\gamma) = F[h]^\times = Z_{\mathrm{GL}_m(F)}(h).$$

Proof. Put

$$\beta = t^{-r}\gamma^d, \qquad B = O[\beta].$$

All eigenvalues of β are units, so its minimal polynomial has coefficients in O and unit constant term. Thus B is a finite, t-adically complete commutative O-algebra, and $\beta^{-1} \in B$.

The quotient B/tB is a finite Artinian k-algebra, all of whose residue fields are k. First choose a d-th root of β in each residue field. This is possible because k is algebraically closed and $d \in O^\times$. The derivative dx^{d-1} is a unit at each root. These roots therefore lift uniquely across the nilpotent radical, and then lift to B by Hensel's lemma, giving

$$h \in B^\times, \qquad h^d = \beta.$$

Choose integers a, b satisfying $ad + br = 1$, and define

$$\pi = t^a \gamma^b h^{-b}.$$

All factors commute, so

$$\pi^d = t^{ad}\gamma^{bd}\beta^{-b} = t^{ad+br} = t,$$

and

$$\pi^r h = t^{ar}\gamma^{br}h^{1-br} = t^{ar}\gamma^{br}h^{ad} = \gamma^{br+ad} = \gamma.$$

The polynomial $X^d - t$ is Eisenstein. Hence $F = L[\pi]$ is the stated field and $O_F = O[\pi]$. The original vector space V becomes an m-dimensional F-vector space.

Let $A = L[\gamma]$. Since $h \in B \subset A$ and $\pi \in A$, we have $F[h] \subset A$. The equality $\gamma = \pi^r h$ gives the reverse inclusion. Regular semisimplicity of γ means that A is an étale L-algebra of dimension N and has a cyclic vector on V. Thus $A = F[h]$ has F-dimension m, and h is a regular semisimple operator on the m-dimensional F-space.

An L-linear operator commuting with γ also commutes with β, h, π. Conversely, an F-linear operator commuting with h commutes with γ. This proves the centralizer equalities. □

4.3 Exact transitivity from cyclic lattices

Proposition 4.3. *For every basic slope of GL_N and every very special endpoint, $Z_{\mathrm{GL}_N}(\gamma)(L)$ acts transitively on $\mathscr{Y}^{\mathrm{reg}}_{K,\tau}(\gamma)$.*

Proof. Suppose that a lattice chain satisfies Proposition 4.1. Periodicity and the shift relation for γ give

$$\beta\Lambda_i = t^{-r}\gamma^d\Lambda_i = \Lambda_i.$$

Thus B preserves every Λ_i, and $h, h^{-1} \in B$ imply $h\Lambda_i = \Lambda_i$. The formula in Lemma 4.2 consequently yields

$$\pi\Lambda_i = \Lambda_{i+ad+br} = \Lambda_{i+1}.$$

The entire chain is therefore determined by one O_F-lattice:

$$\Lambda_i = \pi^i \Lambda_0, \qquad h\Lambda_0 = \Lambda_0.$$

Conversely, any h-stable O_F-lattice Λ gives the original lattice chain $\pi^i\Lambda$, with the prescribed periodicity, quotient dimensions, and shift by γ. This is an equivariant bijection between the original lattice chains and h-stable O_F-lattices. The choices of F, h have been made once from the fixed γ and do not vary with the lattice.

We now compare regularity. The reductive quotient of the original parahoric is $(\mathrm{GL}_m)^d$, and τ cyclically permutes its d factors. The centralizer equations for an element of this cyclic coset determine the coordinates on successive factors, leaving only the equation of commuting with the product around the cycle. Thus the coset element is regular if and only if its product around the cycle is regular in $\mathrm{GL}_m(k)$. Here that product is the reduction of $\beta = \gamma^d/t^r = h^d$ on

$$\Lambda_0/\pi\Lambda_0.$$

Since $h \in O[\beta]$ and $\beta = h^d$, the operators on this quotient satisfy

$$k[\bar{h}] = k[\bar{\beta}].$$

Their centralizers are equal. Hence residual coset regularity is equivalent to $\bar{h}$ being a regular matrix, or equivalently to its minimal polynomial being equal to its characteristic polynomial.

Put $R = O_F[h]$. On a regular lattice, choose a cyclic vector for $\bar{h}$ and lift it to $v \in \Lambda_0$. Nakayama's lemma shows that

$$v, hv, \ldots, h^{m-1}v$$

is an O_F-basis of Λ_0, so $\Lambda_0 = Rv$. The vector v is also a generator of the generic-fiber $A = F[h]$-module. For any two such lattices Rv, Rv', there is a unique $z \in A^\times$ such that $zv = v'$. Therefore

$$z\Lambda_0 = \Lambda_0', \qquad z\Lambda_i = \Lambda_i' \qquad (i \in \mathbb{Z}).$$

Lemma 4.2 identifies z as an element of the centralizer of the original γ. This centralizer consequently acts transitively on all regular lattice chains. Moreover, starting with any generic-fiber cyclic vector and taking the lattice Rv gives an integral companion matrix with cyclic reduction, proving that this regular locus is nonempty. □

This argument uses F-linear algebra internal to the original problem. It does not identify the number of affine Springer fiber components after an arbitrary ramified extension with the original component count. All lifts use only a complete discrete valuation ring, so the argument also applies in mixed characteristic.

4.4 Newton Levi subgroups of GL_n and Sp_{2n}

Proposition 4.4. *Every Newton point of* GL_n *and* Sp_{2n} *satisfies the local transitivity assertion of this section.*

Proof. The Newton Levi subgroup of GL_n is an actual direct product of its slope blocks:

$$M = \prod_j \mathrm{GL}_{a_j}.$$

Each block is basic, and its slope denominator divides its dimension. The affine root system is the disjoint union of those of the factors, and the longest-element lengths add. A τ-stable parahoric is therefore very special if and only if its parahoric in each factor is very

special. Residual coset regularity and the centralizer action also decompose factor by factor. Proposition 4.3 gives transitivity on each factor.

For Sp_{2n}, the positive- and negative-slope spaces are symplectic duals of one another, and the zero-slope space is symplectic. Its actual Newton Levi subgroup is

$$M = \left(\prod_j \mathrm{GL}_{a_j} \right) \times \mathrm{Sp}_{2c}.$$

A factor GL_{a_j} acts by A on the positive-slope space and by A^{-T} on its negative-slope dual. There is no additional determinant condition or finite central constraint. The general linear factors are handled by Proposition 4.3; their denominators $d_j \le a_j \le n$ are invertible in O.

The remaining symplectic factor has slope zero. Since Sp_{2c} is simply connected, its length-zero group Ω is trivial, and the endpoint element τ acts as the identity on this factor. Deleting vertex i from the affine C_c diagram leaves finite type $C_i \times C_{c-i}$, whose longest-element length is

$$i^2 + (c-i)^2.$$

The maximum c^2 occurs only at $i = 0, c$. Deleting further vertices strictly decreases the number of positive roots. Thus the very special endpoints are exactly the two hyperspecial endpoints.

He's hyperspecial local theorem recalled above gives transitivity on regular lattices at the standard hyperspecial endpoint. The L-group automorphism given by conjugation by the symplectic similitude $\mathrm{diag}(tI_c, I_c)$ interchanges the two endpoints, so the other endpoint is also covered. After moving an endpoint coset element into the standard hyperspecial subgroup, choose a stable complete isotropic flag in its reductive quotient to conjugate it into an Iwahori subgroup. This gives the form required by He's theorem. If $c = 0$, there is no symplectic factor to handle.

Taking the product of the transitive actions on the general linear factors and the zero-slope symplectic factor proves the local assertion for every Newton Levi subgroup. Regular semisimple centralizers in GL and Sp are connected maximal tori, so the transporters obtained here belong to the identity-component centralizers. □

4.5 Symplectic similitudes: even multiplier valuations and odd endpoints

To handle all central forms of type C, one needs GSp_{2n}, rather than just Sp_{2n}. On a symplectic space V, write $x^\star$ for the adjoint with respect to the fixed alternating form. For $\Gamma \in \mathrm{GSp}(V)$, its multiplier c satisfies

$$\Gamma^\star \Gamma = cI.$$

The Kottwitz invariant of GSp_{2n} is $r = v_L(c) \in \mathbb{Z}$. If Γ is basic, all slopes of its natural representation are $r/2$.

If r is even, first multiply by the central element $t^{-r/2}$, and then by a central unit, to make the multiplier equal to 1. The required square root of a unit exists by Hensel's lemma. The adjusted element lies in Sp_{2n} and has slope zero. In this case, the length-zero element acts trivially on the affine C_n diagram, and all very special endpoints are the two hyperspecial endpoints. He's hyperspecial local theorem can therefore be applied directly to GSp_{2n}. Lifting a stable isotropic flag of the reduced element conjugates the normalized element into an Iwahori subgroup. The two hyperspecial types are interchanged by conjugation in the symplectic similitude group. This theorem applies directly to the entire regular locus in GSp/P_K, including all central components. Central multiplication adjusts the associated τ at the same time and preserves the local problem for regular points.

Henceforth suppose that r is odd. By multiplying by a suitable central unit, we can, once and for all, arrange that

$$c = -t^r, \qquad \Gamma^\star = -t^r\Gamma^{-1}.$$

A central unit belongs to every connected parahoric subgroup, so this normalization changes neither the relevant fiber nor residual coset regularity.

The nontrivial length-zero action sends vertex i of the affine C_n diagram to $n-i$. The maximal τ-stable spherical subsets are obtained by deleting one vertex orbit; deleting further orbits strictly decreases the longest-element length. Deleting $\{j, n-j\}$, where $0 \le j < n/2$, leaves type

$$C_j \times A_{n-2j-1} \times C_j,$$

with length

$$f(j) = 2j^2 + \frac{(n-2j-1)(n-2j)}{2}.$$

This is a strictly convex quadratic polynomial, so it suffices to compare the two ends of the allowable interval.

If $n = 2m$, one may also delete the single middle vertex m. The resulting type $C_m \times C_m$ has length $2m^2$. The largest value among the other candidates is at most

$$\max\{f(0), f(m-1)\} = \max\{m(2m-1), 2(m-1)^2 + 1\} < 2m^2.$$

Thus the unique very special subset deletes the middle vertex. If $n = 2m+1 \ge 3$, then

$$f(0) = m(2m+1) > 2m^2 = f(m),$$

so the unique very special subset deletes the endpoint pair $\{0, n\}$, leaving type A_{n-1}. For $n = 1$, this subset is empty, as expected from $\mathrm{GSp}_2 = \mathrm{GL}_2$.

4.6 Self-dual indexed two-step lattice chains

In the natural representation, both of the preceding kinds of odd-valuation endpoints are described by two-step chains satisfying

$$\Lambda_{i+2} = t\Lambda_i, \qquad \dim_k(\Lambda_i/\Lambda_{i+1}) = n, \qquad \Gamma\Lambda_i = \Lambda_{i+r}.$$

If Λ^* denotes the lattice dual with respect to the symplectic form, they also satisfy

$$\Lambda_i^* = t^{-q}\Lambda_{a-i},$$

where a is fixed, while the integer q may vary with the flag point. The indexing can be chosen with $a = 1$ for even n and $a = 0$ for odd n.

For even n, the reference middle lattice satisfies

$$t\Lambda_0^* \subset \Lambda_0 \subset \Lambda_0^*, \qquad \Lambda_1 = t\Lambda_0^*.$$

Both steps have length n, and the duality formula above holds with $q = 1, a = 1$. For odd n, the reference endpoint chain can be chosen with Λ_0 self-dual and with $\Lambda_1^* = t^{-1}\Lambda_1$, in which case $q = 0, a = 0$. Applying an arbitrary similitude changes only the common exponent q, without changing a. The unit part of its multiplier has no effect on equalities between lattices.

These indexed chains represent precisely the points of GSp/P_K. Indeed, if g fixes the lattice at every index, then $v_L(\det g) = 0$. Since

$$\det g = \mathrm{sim}(g)^n,$$

its multiplier has valuation zero, and its Kottwitz invariant is zero. The elements fixing this facet and having zero Kottwitz invariant are exactly the connected parahoric subgroup. Conversely, this connected parahoric fixes the entire indexed chain. Equality of lattice chains therefore loses no additional connected-component label.

4.7 Residual regularity implies cyclicity in both odd-valuation cases

Put

$$\beta = t^{-r}\Gamma^2.$$

It preserves each lattice, and its reduction on Λ_0/Λ_1 is the product around the cycle of the two-step coset.

Lemma 4.5. *If the residual coset element of this* GSp *lattice chain is regular, then $\bar{\beta}$ is a cyclic matrix on the n-dimensional quotient.*

Proof. If n is even, the semisimple part of the reductive quotient is $\mathrm{Sp}_n \times \mathrm{Sp}_n$, and τ interchanges the two factors. The central similitude factor adds the same one dimension to every centralizer. Solving the two commutation equations shows that coset regularity is equivalent to ordinary regularity of the product around the cycle in Sp_n.

An ordinary regular element of a symplectic group is cyclic in its natural representation. To see this, decompose according to the eigenvalues of its semisimple part. Reciprocal eigenvalue pairs different from ± 1 give general linear factors, while the $+1$ and -1 eigenspaces give symplectic factors. Regularity forces the unipotent part to be regular unipotent in every factor. Each natural eigenspace of a general linear factor has one Jordan block, and the natural representation of each symplectic factor also has one Jordan block of even size. Thus the full matrix has exactly one Jordan block for each eigenvalue and is cyclic.

If n is odd, the reductive quotient can be written as $\mathrm{GL}_n \times \mathbb{G}_m$, and the length-zero action is

$$(A, c) \longmapsto (cA^{-T}, c).$$

After separating the fixed central similitude direction, this is the transpose-inverse twisted coset of GL_n. In an isotropic decomposition, take the reference representative

$$\dot{\tau} = \begin{pmatrix} 0 & tI \\ I & 0 \end{pmatrix}, \qquad \mathrm{sim}(\dot{\tau}) = -t.$$

Representing a residual element by $\mathrm{diag}(A, A^{-T})$, its product around the cycle is AA^{-T}, the asymmetry operator of the corresponding nondegenerate bilinear form. Every odd r differs from this representative by a central translation, so it gives the same finite-dimensional problem.

Suppose that the $+1$ and -1 eigenspaces of the semisimple asymmetry operator have dimensions n_+, n_-, respectively, and that one side of a reciprocal eigenvalue pair $\{\lambda, \lambda^{-1}\}$, with $\lambda \neq \pm 1$, has dimension n_λ. The identity component of the centralizer of the twisted semisimple element is

$$\mathrm{SO}_{n_+} \times \mathrm{Sp}_{n_-} \times \prod_{\{\lambda,\lambda^{-1}\}} \mathrm{GL}_{n_\lambda}.$$

Indeed, the bilinear form is symmetric on the $+1$ space, alternating on the -1 space, and pairs the reciprocal spaces nondegenerately. Hence n_- is even. Since the total dimension n is odd, n_+ must be odd. The sum of the ranks of these factors is always

$$\frac{n_+ - 1}{2} + \frac{n_-}{2} + \sum n_\lambda = \frac{n-1}{2},$$

which is the smallest possible centralizer dimension in this twisted coset. Coset regularity therefore forces the unipotent part to be regular in each factor.

The natural Jordan type of a regular unipotent element is (n_+) in the odd-dimensional orthogonal factor and (n_-) in the symplectic factor, while each general linear natural space has a single block. When the twisted element is squared to obtain the asymmetry operator,

its unipotent part is squared. Since 2 is invertible, this does not change its Jordan block sizes. Thus $\bar{\beta}$ still has exactly one Jordan block for each eigenvalue.

The crucial point is that n_+ is odd, so the regular but noncyclic type $(2a-1,1)$ that can occur in an even-dimensional orthogonal group does not arise. The disconnectedness of the full orthogonal centralizer changes neither its dimension nor the unipotent calculation in its identity component. The classical Jordan types used here are valid in the characteristic range of this manuscript. This proves the lemma. □

4.8 An integral square root compatible with the adjoint

The normalization $\Gamma^\star = -t^r\Gamma^{-1}$ gives

$$\beta^\star = \beta^{-1}.$$

Thus $B = O[\beta]$ is a finite complete commutative algebra, $\beta \in B^\times$, and $\star$ preserves B. It sends the local factor with residual eigenvalue λ to the local factor with residual eigenvalue λ^{-1}.

Define $\epsilon \in B$ to be -1 on the local factor with residual eigenvalue -1, and $+1$ on every other factor. Idempotents in the complete algebra lift uniquely from the reduction, so

$$\epsilon^2 = 1, \qquad \epsilon^\star = \epsilon.$$

On distinct reciprocal factors, choose reciprocal residual square roots. On the $+1$ factor choose 1, and on the -1 factor choose an element $i \in k$ satisfying $i^2 = -1$. Hensel lifting, with unit derivative $2h$, gives

$$h \in B^\times, \qquad h^2 = \beta, \qquad h^\star = \epsilon h^{-1}.$$

The last equality follows from uniqueness of the lift of the prescribed residual roots: on the -1 factor, both $h^\star$ and $-h^{-1}$ reduce to i and have square β^{-1}; on the other factors, the roots were chosen reciprocally.

Choose integers a_0, b_0 satisfying $2a_0 + b_0 r = 1$. Since r is odd, b_0 is odd. Define

$$\pi = t^{a_0}\Gamma^{b_0}h^{-b_0}.$$

Here we reuse only the commutative-algebra calculation in Lemma 4.2 with $d = 2$. It requires only that 2 be invertible, not that $p \nmid (2n)!$. We obtain

$$\pi^2 = t, \qquad \Gamma = \pi^r h.$$

Since b_0 is odd, the adjoint identities also yield

$$\pi^\star\pi = (-1)^{b_0}\epsilon^{-b_0}t^{2a_0+b_0r} = -\epsilon t,$$

and hence

$$\pi^\star = -\epsilon\pi.$$

Therefore $F = L[\pi]$ is a totally ramified quadratic field inside the original operator algebra, $O_F = O[\pi]$, and all relevant lattice chains satisfy

$$\Lambda_i = \pi^i\Lambda_0.$$

A regular semisimple GSp element has distinct eigenvalues in its natural representation: a repeated reciprocal eigenspace would give a nontoral general linear centralizer, and an eigenspace satisfying $\lambda^2 = c$ would give a nontoral symplectic centralizer. Consequently,

$$A = L[\Gamma] = F[h]$$

is an étale L-algebra of dimension $2n$, and is also the full commutative centralizer algebra on the n-dimensional F-space. Put

$$R = O_F[h].$$

Since $\epsilon \in O[\beta] = O[h^2]$, $h^{-1} \in R$, and $h^\star = \epsilon h^{-1}$, $\pi^\star = -\epsilon\pi$, the integral algebra R is preserved by the adjoint. The adjoint need not act in the same way on F in all factors of A; the proof requires only stability of R.

Lemma 4.5, together with $k[\bar{h}] = k[\bar{\beta}]$, shows that $\bar{h}$ is cyclic. Lifting a cyclic vector v gives

$$\Lambda_0 = Rv.$$

In particular, the multiplier ring of Λ_0 in A is exactly R: if $a\Lambda_0 \subset \Lambda_0$, then $av \in Rv$, and because v is a generic-fiber generator, $a \in R$. If the lattices are equal, then $a \in R^\times$.

4.9 Integral norm correction and GSp transitivity

Proposition 4.6. *For every basic Kottwitz class of GSp_{2n}, including the classes with odd multiplier valuation, the identity-component centralizer torus acts transitively on the regular points at every very special endpoint.*

Proof. The even classes were handled in Subsection 4.5. Suppose that two regular points in an odd class are represented by chains Λ_i, Λ_i', whose duality exponents are q, q', respectively. Both initial lattices are rank-one cyclic R-modules, so there exists

$$z \in A^\times, \qquad z\Lambda_0 = \Lambda_0'.$$

Since z commutes with π, it carries the entire indexed chain to the other one. The adjoint also preserves A, so $z^\star \in A$. Using

$$(z\Lambda_0)^* = (z^\star)^{-1}\Lambda_0^*$$

and $\Lambda_0^* = t^{-q}\pi^a\Lambda_0$, $(\Lambda_0')^* = t^{-q'}\pi^a\Lambda_0'$, and canceling the commuting factor π^a, gives

$$z^\star z\Lambda_0 = t^{q'-q}\Lambda_0.$$

Hence

$$u = t^{-(q'-q)}z^\star z \in R^\times, \qquad u^\star = u.$$

Membership $u \in R^\times$ follows from the multiplier-ring property just proved.

Every $\star$-fixed unit $u \in R^\times$ has a $\star$-fixed inverse square root. More explicitly, choose compatible residual square roots of u^{-1} on the orbits of the residual maximal ideals of R. At a fixed ideal the residue field is k, and $\star$ acts as the identity on that field, so both square roots are fixed. For a pair of exchanged ideals, carry the root on one side to the other. Lift successively across the nilpotent radical and through the complete topology by Hensel's lemma. The derivative is a unit, and uniqueness of the lift of the prescribed residual roots gives

$$w \in R^\times, \qquad w^\star = w, \qquad w^2 = u^{-1}.$$

Set $g = zw$. Since w is a unit automorphism of every $\pi^i\Lambda_0$, the element g still carries the original entire chain to the other one, while

$$g^\star g = w^\star z^\star zw = t^{q'-q}uw^2 = t^{q'-q}I.$$

Thus $g \in \mathrm{GSp}_{2n}(L)$. It lies in $A^\times$, so it commutes with Γ. Subsection 4.6 showed that the stabilizer of the indexed chain is exactly the connected parahoric subgroup. Consequently, g is a transporter between the original two points of GSp/P_K. The derived group of GSp is simply connected, and a regular semisimple centralizer is a connected maximal torus. Hence g belongs to the required identity-component centralizer. □

The entire odd-class argument requires only that O be complete and that 2 be invertible. Even when F/L is ramified in mixed characteristic, no additional geometric transitivity theorem over F is invoked.

4.10 All central forms of types A and C and all Newton points

The actual Newton Levi subgroup of GSp_{2n} is

$$M = \left(\prod_j \mathrm{GL}_{a_j} \right) \times \mathrm{GSp}_{2c},$$

where for $c = 0$ the last factor is $\mathbb{G}_m$. If the multiplier of the last factor is c_0, then a general linear factor acts by A on the space of positive slope relative to the central slope and by $c_0 A^{-T}$ on its paired space. This gives independent direct-product coordinates without any determinant or finite central constraint. Consequently, Propositions 4.3 and 4.6, together with the factorwise decomposition of regularity and maximal longest-element length, prove local transitivity for all Newton points of GSp.

The adjoint groups of types A and C are obtained, respectively, from

$$\mathrm{GL}_a \longrightarrow \mathrm{PGL}_a, \qquad \mathrm{GSp}_{2c} \longrightarrow \mathrm{PSp}_{2c},$$

both with split scalar torus kernel. The proof by lifting fixed cosets in Section 3 descends the local property to these adjoint groups, including every Kottwitz class. Applying the natural central extension and graph isogeny from that section to every actual Newton Levi subgroup then gives all central forms of types A and C: forms with simply connected derived group, adjoint forms, extensions by central tori, and central quotient gluings between different factors.

In the finite-cover step, only transporters whose Kottwitz invariants already belong to the image of the source are lifted; the fixed element is first lifted through a split torus kernel. Thus the argument neither assumes that an odd similitude class lifts directly to Sp nor omits any finite central grading.

Combining this with the reduction of Section 2, every top-dimensional path has $n_{p,\gamma} = 1$, and hence all these groups satisfy

$$\#\left(Z_G(\gamma)(L) \backslash \Sigma^{\mathrm{top}} Y_\mu(\gamma) \right) = \dim V_\mu(\lambda_b).$$

Here the full Kottwitz information in λ_b is retained; it has not been replaced by the Newton vector, which may be nonintegral.

5 Ordinary and Vector Classes in Types B and D

This section proves the local assertion at every Newton point for the actual split group SO_N. Here the *vector class* means the nonzero Kottwitz class of SO_N; it is distinct from the half-spin classes of the even-dimensional adjoint group, which are treated separately in the next section. Let $\mathcal{O}$ be the original discrete valuation ring, let t be a uniformizer, and put $L = \mathrm{Frac}(\mathcal{O})$. Its residue field k is algebraically closed. Apart from the trivial-group and torus cases SO_1 and SO_2, the characteristic hypothesis of this section ensures that $2 \in \mathcal{O}^\times$.

5.1 The actual Newton Levi and the required local assertion

In a standard hyperbolic basis, decompose the standard representation into the positive, negative, and zero weight spaces of the Newton cocharacter. The matrices that preserve the

quadratic form and commute with this cocharacter act by A_j on each positive weight space, by ${}^tA_j^{-1}$ on the corresponding negative weight space, and by a special orthogonal matrix on the zero weight space. The product of the determinants on every positive–negative pair is one. Consequently,

$$M_\nu \simeq \prod_j GL_{a_j} \times SO_d$$

is an actual direct product: there is neither a hidden determinant constraint nor a finite central quotient. The quadratic space of weight zero consists of standard hyperbolic pairs, together with the standard one-dimensional space in odd dimension, and remains split. The distinction between the two end-node embeddings of type A Levi subgroups in an even orthogonal group does not change this direct-product conclusion. We interpret SO_0 and SO_1 as trivial factors, and $SO_2 \simeq \mathbb{G}_m$.

Local transitivity on regular points for the GL factors has already been proved by the cyclic-lattice argument. Thus it remains to prove the following assertion for a basic element $\gamma \in SO_N(L)$: at every very special endpoint $P\tau$ associated with γ, the points of the regular part of the partial affine Springer fiber form a single orbit under

$$T(L), \qquad T = Z_{SO_N}(\gamma)^\circ.$$

Throughout this section we use this identity-component torus; we do not assume that the full group centralizer is connected. Once the assertion is proved, He's local component theorem gives $n_{p,\gamma} = 1$ for every actual top-dimensional reduction path. The path-counting formula then gives the conclusion for SO_N.

5.2 All very special endpoints for the ordinary and vector classes

First suppose that $N \geq 3$. In the standard coordinates $e_1, \ldots, e_r$, the cocharacter and coroot lattices of the actual groups SO_{2r+1} and SO_{2r} are

$$X_*(T_0) = \mathbb{Z}^r, \qquad Q^\vee = \left\{ (a_i) \in \mathbb{Z}^r : \sum_i a_i \equiv 0 \pmod 2 \right\}.$$

Hence $\pi_1(SO_N) = \mathbb{Z}/2$. Its nonzero class is represented by e_1, whose action on the affine diagram is

$$B_r : \ (0\ 1), \qquad D_r : \ (0\ 1)(r-1\ r).$$

For $B_1 = A_1$, it exchanges the two affine vertices. For $D_2 = A_1 \sqcup A_1$, it simultaneously exchanges the two vertices in each of the two affine A_1 components. In particular, the affine D_2 diagram must not be treated as a connected three-vertex diagram.

The very special condition compares the numbers of positive roots of stable spherical root subsets. The action of the ordinary class is the identity. For B_r, the maximum is r^2, attained by deleting 0 or 1; for D_r, it is $r(r-1)$, attained by deleting any one of the four end vertices. These are precisely the hyperspecial types. The diagram B_1 has two such vertices. For D_2, one deletes one vertex from each affine A_1 component, giving four possibilities, each with two positive roots.

For the nonzero class, every stable spherical root subset is contained in the root subset obtained by deleting one whole vertex orbit. Further deletion of vertices strictly decreases the number of positive roots, so it suffices to compare deletions of a single orbit. In B_r, for $r \geq 2$, deleting $\{0, 1\}$ leaves $(r-1)^2$ positive roots; deleting a fixed vertex j, with $2 \leq j \leq r$, leaves

$$f_B(j) = j(j-1) + (r-j)^2.$$

This function is strictly convex. Comparing its two endpoints shows that the unique maximum is attained at $j = r$, where its value is $r(r-1)$, and the remaining root system is D_r. Here $D_2 = A_1 \sqcup A_1$ and $D_3 = A_3$. In D_r, for $r \geq 4$, deleting either $\{0,1\}$ or $\{r-1, r\}$ leaves $(r-1)(r-2)$ positive roots, with remaining root system D_{r-1}. Deleting a fixed interior vertex gives

$$f_D(j) = j(j-1) + (r-j)(r-j-1), \qquad 2 \leq j \leq r-2.$$

Its maximum, $2 + (r-2)(r-3)$, is smaller than the end-orbit value by $2r - 6 > 0$. Thus only the two end orbits give the maximum. The affine diagram of $D_3 = A_3$ is a square, and the vector action exchanges the two pairs of opposite vertices. Deleting either orbit leaves $A_1 \sqcup A_1$, and both choices are maximal. For B_1 and D_2, the only stable spherical subset is empty, and the connected reductive quotients are respectively SO_2 and $SO_2 \times SO_2$.

The hyperspecial models for the ordinary class are self-dual lattices. In even dimension, one may also obtain a t-modular lattice satisfying $\Lambda^* = t^{-1}\Lambda$. Replacing the quadratic form B by $t^{-1}B$ makes such a lattice self-dual; moreover, $SO(B) = SO(t^{-1}B)$, and the generic fiber remains split. This rescaling includes all the even orthogonal hyperspecial types.

The lattice models for maximal endpoints in the nonzero class have the uniform form

$$\Lambda = U_{\mathcal{O}} \perp W_{\mathcal{O}}, \qquad B|_\Lambda = B_0 \perp tB_1,$$

where B_0 and B_1 are unimodular, with dimensions as follows.

Group	$(\dim U, \dim W)$	Root system of the connected reductive quotient
SO_{2r+1}, $r \geq 1$	$(1, 2r)$	D_r; a torus for $r = 1$
SO_{2r}, $r \geq 3$	$(2r-2, 2)$ or $(2, 2r-2)$	D_{r-1}
SO_4	$(2, 2)$	A torus

These lattices exist in the split generic fiber: an even-dimensional unimodular space is hyperbolic, and multiplying its form by t again gives a hyperbolic space. For each fixed row of the table, the lattices of that model form one $SO_N(L)$-orbit. First match them by an orthogonal isometry, and then correct the determinant to one using a reflection in a unimodular vector that preserves the target lattice.

The full lattice stabilizer has reductive quotient

$$S\big(O(B_0) \times O(B_1)\big),$$

in which the determinant signs of the two factors agree. Its kernel is connected unipotent, and the connected parahoric P has reductive quotient $SO(B_0) \times SO(B_1)$. Consequently the full stabilizer modulo P has exactly two cosets; the nontrivial coset has signs $(-1, -1)$. This sign is precisely the Kottwitz class for SO. Indeed, all units of $\mathcal{O}$ are squares, and the product of one reflection in a vector of unit norm and one reflection in a vector of norm t times a unit has determinant one and spinor norm of odd valuation. Thus each such lattice has exactly two connected-parahoric flag decorations, and both must be handled at the end.

5.3 The ordinary regular lattice lemma over an auxiliary complete DVR

Lemma 5.1 (Ordinary regular lattice lemma)**.** *Let R be a complete DVR with algebraically closed residue field and fraction field E, satisfying the characteristic hypothesis of this section. The ring R is allowed to be a ramified extension of the original valuation ring. Let (V, B) be a split quadratic E-space, and let $x \in SO(V)(E)$ be regular semisimple. Suppose that the*

self-dual lattices L_1, L_2 are preserved by x, and that $\bar{x}$ is regular in the special orthogonal group of each lattice. Then there exists

$$h \in Z_{SO(V)}(x)^\circ(E), \qquad hL_1 = L_2.$$

For any such lattice L, the stabilizer of L in the generic identity-component centralizer torus has an intrinsically defined smooth affine R-model J. Its special fiber contains $Z_{SO(L/\varpi L)}(\bar{x})^\circ$. If a finite-order semilinear automorphism preserves B, x, L, then the construction of J is equivariant for this automorphism.

Proof. Since x preserves the lattices, its characteristic polynomial lies in $R[X]$ and has unit constant term. The pairwise coprime factors corresponding to the residual eigenvalues have unique Hensel lifts. The associated Bézout idempotents lie in $R[x]$, and therefore decompose both lattices into intrinsic spectral clusters $L_{i,\lambda}$. The quadratic form pairs the λ-cluster with the λ^{-1}-cluster; the lattices in a non-self-reciprocal pair are dual to each other.

If $\lambda \neq \pm 1$, regularity makes $\bar{x}$ a regular GL_d-matrix on that cluster, so it has a cyclic vector. Lift such a residual cyclic vector to v_i. Nakayama's lemma gives

$$L_{i,\lambda} = R[x]v_i.$$

On the generic fiber, take the $E[x]$-linear map sending v_1 to v_2, and take its inverse transpose on the dual cluster. The resulting map has determinant one and belongs to the generic identity-component centralizer. The stabilizer model on this cluster is the unit group $A^\times$ of the finite free algebra $A = R[x|_{V_\lambda}]$. It is the unit open subset of the underlying affine space, hence is smooth; its special fiber is the corresponding GL-centralizer.

The residual $+1$ and -1 clusters are individually unimodular. Since 2 is invertible and every unit has a square root by Hensel's lemma, their quadratic forms are split over R. On the $+1$ cluster take $y = x$, and on the -1 cluster take $y = -x$. The negative cluster has even dimension: the determinants on the other pairs of reciprocal spectral clusters are one, and $\det \bar{x} = 1$ forces the dimension of the negative cluster to be even. Thus both choices of y belong to the appropriate special orthogonal group. The operator $y + 1$ is a lattice automorphism, so the Cayley transform

$$X = (y - 1)(y + 1)^{-1}$$

is integral and skew-adjoint. The inverse relation $y = (1 + X)(1 - X)^{-1}$ shows that on the integral model, special fiber, and generic fiber, the group centralizer of y equals the Lie centralizer of X. The element $\bar{X}$ is regular. The zero Lie element of SO_2 is also regular, because this group is a torus. This step does not mistake the regular even orthogonal unipotent partition $(2a - 1, 1)$ for a cyclic matrix.

We now give the integral consequence of the finite-dimensional Kostant input in detail. For split $H = SO_d$, the coordinates of the Lie quotient $\chi : \mathfrak{h} \to \mathfrak{c}$ in odd dimension are the independent even-degree characteristic coefficients. In even dimension they are the coefficients of degrees $2, 4, \ldots, 2a-2$, together with the degree-a Pfaffian. The last characteristic coefficient in even dimension is the square of the Pfaffian, so the characteristic polynomial cannot replace the full set of quotient coordinates. All these coordinates are explicit polynomials over R.

The standard finite-dimensional regular-quotient theorem states that, on the allowed geometric fibers, χ is smooth on the regular locus, every geometric regular fiber is a single H-orbit, and the regular Lie centralizer is smooth of dimension $\operatorname{rk} H$. The locus where the Jacobian of the full quotient polynomials has full rank is precisely the regular locus on both geometric fibers. If an element has regular reduction, the relevant full-rank minor is a unit. Thus χ is smooth on its integral model at that element.

Consider the action map

$$H \times \mathfrak{h}_{\mathrm{reg}} \longrightarrow \mathfrak{h}_{\mathrm{reg}} \times_{\mathfrak{c}} \mathfrak{h}_{\mathrm{reg}},$$
$$(g, X) \longmapsto (\mathrm{Ad}(g)X, X).$$

At a geometric regular point, $\ker d\chi_X = [\mathfrak{h}, X]$, so its differential $(U, V) \mapsto ([U, X] + V, V)$ is surjective. The source and target are both smooth over R; hence this action map is smooth. The single-orbit property also ensures that it is surjective on every geometric fiber. Therefore the transporter between two integral regular Lie elements with the same full quotient invariants is a smooth R-scheme whose special fiber has a k-point. Completeness and smooth Hensel lifting give an R-point. Pulling back the same map along (X, X) also shows that the integral regular Lie centralizer is smooth.

To compare the ± 1 clusters of the two self-dual lattices, first fix a volume form in their common generic quadratic space that is compatible with the standard split discriminant. It is a generator of the integral determinant lattice for each self-dual lattice: the norm of the volume is a unit, so its valuation is zero. Choose integral split isometric coordinates respecting this common orientation, adjusting by an integral reflection of determinant minus one if necessary. The two generic Cayley matrices are conjugate by an SO-isometry, so their full quotient invariants coincide in E, including the Pfaffian in even dimension. Since both sets of invariants lie in R, they also coincide in R. The smooth transporter above gives an integral SO-conjugating map. Viewed back in the common space, this map commutes with x.

This conjugating map belongs to the generic identity-component centralizer. Indeed, for a regular semisimple Lie element, the centralizers on its nonzero reciprocal spectral pairs are GL tori; its zero spectral cluster has dimension at most two, and the corresponding SO-centralizer is trivial or a torus. Hence the generic Lie centralizer is connected. Use this integral Lie centralizer model as the stabilizer model on each ± 1 cluster. The product of all the cluster models is J; its generic fiber is exactly $T = Z_{SO(V)}(x)^\circ$, and $J(R)$ consists exactly of the elements of $T(E)$ preserving L.

The identity component of the finite group centralizer has determinant one on each ± 1 cluster, and acts by inverse transposes on the other pairs of spectral clusters. It is therefore contained in J_k. Smoothness gives

$$J(R) \twoheadrightarrow J(k) \supset Z_{SO(L/\varpi L)}(\bar{x})^\circ(k).$$

The spectral-cluster idempotents, Cayley operators, and centralizer models are all defined intrinsically from x, B, L. A semilinear automorphism can at most permute these factors, so the construction is equivariant. The product of the conjugating maps on all the clusters proves the asserted transitivity on lattices. □

Lemma 5.1 immediately handles all hyperspecial types in the ordinary class. For a t-modular lattice, first rescale the quadratic form as in Subsection 5.2. No affine Springer transitivity theorem for ramified groups is invoked here.

5.4 The quadratic self-dual lift of the vector class and ordinary regularity

Let γ be a regular semisimple element in the nontrivial stabilizer coset at an endpoint listed in Subsection 5.2. Put

$$F = L(\pi), \qquad \pi^2 = t, \qquad \sigma(\pi) = -\pi.$$

For any corresponding regular lattice, define the intrinsic lift

$$\widetilde{\Lambda} = \Lambda_{\mathcal{O}_F} + \pi\Lambda^*_{\mathcal{O}_F} = U_{\mathcal{O}_F} \perp \pi^{-1}W_{\mathcal{O}_F}.$$

It is self-dual and σ-stable. Moreover, $\mathcal{O}_F \cap L = \mathcal{O}$ and $\pi^{-1}\mathcal{O}_F \cap L = \mathcal{O}$, so $\widetilde{\Lambda}^\sigma = \Lambda$. The residual descent operator S acts by $+1$ on $\bar{U}$ and by -1 on $\overline{\pi^{-1}W}$. Since $\dim W$ is even, $S \in SO_N(k)$.

If γ preserves Λ, it also preserves Λ^*. In a basis adapted to U, W, its block from W to U is divisible by t; after passing to the basis $U, \pi^{-1}W$, both off-diagonal blocks are divisible by π. Thus the residual element $\Gamma = \bar{\gamma}$ is the direct sum of the two original finite orthogonal actions, and commutes with S.

We explain why a regular parahoric coset gives an ordinarily regular element $\Gamma \in SO_N(k)$. For an element of determinant minus one in O_{2a}, its semisimple $+1$ and -1 eigenspaces both have odd dimension. Its identity-component semisimple centralizer is the product of the two odd orthogonal factors and some GL factors, and has rank $a-1$. The centralizer has its minimum dimension precisely when the unipotent part is regular in every factor. Thus the two distinguished spectral clusters have single odd blocks (p) and (q), while the GL blocks on all other reciprocal spectral pairs are cyclic. For $a = 1$, an element of determinant minus one is simply a reflection.

For odd-dimensional SO_{2r+1}, the small factor is a line on which γ acts by -1. Adding this line changes the negative cluster from (q) to $(q, 1)$, the regular unipotent partition for an even orthogonal group; the positive cluster remains a regular odd orthogonal block. In even dimension the small factor is a reflection plane, adding one line to each of the positive and negative clusters and giving $(p, 1)$ and $(q, 1)$. The other GL blocks remain cyclic. Hence Γ is ordinarily regular in both cases. When $p = q = 1$, the resulting SO_2 factors are tori, whose identity unipotent elements are regular. This includes the boundary cases SO_3, SO_4, and SO_6.

5.5 Correcting nondegenerate lines within the finite identity centralizer

Lemma 5.2 (Invariant-line transitivity)**.** *If $u \in SO_{2a}(k)$ is regular unipotent, then $Z_{SO_{2a}}(u)^\circ(k)$ acts transitively on the u-invariant nondegenerate lines.*

Proof. If $a = 1$, then SO_2 is a split torus. In a hyperbolic plane, both coordinates of a nondegenerate line are nonzero. The torus action and the existence of square roots in k give the required transitivity.

For $a > 1$, the regular partition is $(2a-1, 1)$. Decompose the space orthogonally into a nondegenerate odd block and a trivial nondegenerate line ke, normalized by $(e, e) = 1$. The fixed vector v of the odd block is isotropic, so the invariant nondegenerate lines are precisely $k(e + cv)$. Put $X = (u-1)(u+1)^{-1}$; then $Xe = Xv = 0$. Define

$$Y(z) = (e, z)v - (v, z)e, \qquad T_c = 1 + cY + \frac{c^2}{2}Y^2.$$

The operator Y is skew-adjoint. The relations $Xe = Xv = 0$ give $[X, Y] = 0$, and $Y^3 = 0$. Direct multiplication gives $T_c^* T_c = 1$ and $T_c T_d = T_{c+d}$. These elements form a one-parameter unipotent group, so they have determinant one and lie in the identity-component centralizer. Moreover, $T_c e = e + cv$. This proves transitivity on the listed lines. □

Now let Γ be the ordinarily regular element obtained in Subsection 5.4. In odd dimension, the $+1$ space of the descent involution is the distinguished nondegenerate line on which $\Gamma = -1$; its remaining space has negative sign. Apply Lemma 5.2 within this even orthogonal negative cluster. In even dimension, the distinguished plane of the descent involution carries a reflection for Γ. It is therefore the orthogonal sum of one nondegenerate line with $\Gamma = +1$ and another with $\Gamma = -1$. Apply Lemma 5.2 separately in these two even orthogonal spectral clusters. No correction is needed on the other reciprocal spectral clusters.

Consequently, any two involutions S, S' that commute with the fixed Γ, have the same endpoint type, and have the precise distinguished line or plane described above are conjugate under $Z_{SO_N}(\Gamma)^\circ(k)$. This assertion specifies the action of Γ on the distinguished spaces; it cannot be deduced solely from the dimensions of the involution eigenspaces.

5.6 Full residual correction, congruence averaging, and flag decorations

Take two regular lattices Λ_1, Λ_2 of the same endpoint type. By Subsection 5.4, their self-dual lifts both have ordinarily regular reductions. Lemma 5.1, applied over $\mathcal{O}_F$, gives

$$h \in T(F), \qquad h\widetilde{\Lambda}_1 = \widetilde{\Lambda}_2.$$

Put $a = h^{-1}\sigma(h)$. It stabilizes $\widetilde{\Lambda}_1$, hence belongs to $J(\mathcal{O}_F)$ from Lemma 5.1, and satisfies $a\sigma(a) = 1$. The model J is σ-stable: the integral characteristic polynomial of γ is defined over L, and therefore belongs to $\mathcal{O}[X]$; its Hensel spectral clusters and idempotents are uniquely defined over $\mathcal{O}$. The Cayley transforms, duality forms, oriented SO factors, and centralizer models are all intrinsic.

Transport the descent operator of the target lattice to the first lifted lattice. Its residual operator is

$$S' = \bar{a}S.$$

It commutes with Γ, and its distinguished line or plane has the same dimension and the same Γ-action as that of S. By Subsection 5.5, choose $\bar{b} \in Z_{SO_N}(\Gamma)^\circ(k)$ such that

$$\bar{b}^{-1}S'\bar{b} = S, \qquad \bar{b}^{-1}\bar{a}\,\sigma(\bar{b}) = 1.$$

Lemma 5.1 ensures that this full finite identity-component centralizer is contained in J_k. Smoothness therefore lifts $\bar{b}$ to $b \in J(\mathcal{O}_F)$. Replace h by hb; its descent discrepancy is then trivial modulo π. Here we have neither identified the entire finite centralizer with a connected group nor assumed that its H^1 is trivial.

The remaining discrepancy lies in the congruence kernel. If it is trivial modulo π^i, its image A in

$$J^i/J^{i+1} \simeq \mathrm{Lie}(J_k) \otimes_k \pi^i\mathcal{O}_F/\pi^{i+1}\mathcal{O}_F$$

satisfies $A + \sigma(A) = 0$. Take a correction whose image is $A/2$. The image of the new discrepancy is

$$A - A/2 + \sigma(A/2) = 0.$$

Smoothness allows these corrections to be lifted successively, and completeness gives their π-adic limit. The resulting element $h_0 \in T(L)$ is fixed by σ and still takes $\widetilde{\Lambda}_1$ to $\widetilde{\Lambda}_2$. Taking fixed lattices gives $h_0\Lambda_1 = \Lambda_2$.

Finally, consider the two connected-parahoric flag decorations on each lattice. If the decoration transported by h_0 is incorrect, multiply on the left by γ at the target. Since γ belongs to the nontrivial coset of the full lattice stabilizer, it exchanges exactly these two decorations. Moreover, γ is semisimple and belongs to an L-maximal torus containing it; this torus is $T = Z_{SO_N}(\gamma)^\circ$. Thus the final correction still lies in $T(L)$. It follows that the regular part of the partial fiber at every very special endpoint of the vector class is transitive under the identity-component centralizer.

5.7 Conclusion for the actual special orthogonal groups

The ordinary class is handled by Lemma 5.1 and t-modular rescaling; the nonzero vector class is handled by Subsections 5.4–5.6. The comparisons valid in all ranks in Subsection 5.2, together with the separate calculations for B_1, D_2, D_3, ensure that no endpoint has been

omitted. The group SO_1 is trivial and SO_2 is a torus. Combined with the actual $\prod GL \times SO$ Newton Levi decomposition of Subsection 5.1, these results give the local transitivity required by every actual top-dimensional reduction path.

To explain the passage from points to components, choose in each top-dimensional component a point that does not belong to any other top-dimensional component. The locally finite type property, with the corresponding perfection in mixed characteristic, guarantees the existence of such an open subset. Transitivity on points then implies transitivity on top-dimensional components. He's local component correspondence consequently gives $n_{p,\gamma} = 1$ for every path, and hence

$$\begin{aligned} \#\big(Z_{SO_N}(\gamma)(L)\backslash \Sigma^{\mathrm{top}} Y_\mu(\gamma)\big) &= \sum_{p\in\mathcal{R}} 1 \\ &= |\mathcal{R}| \\ &= \dim V_\mu(\lambda_b). \end{aligned}$$

This proves the assertion at all Newton points for the actual split group SO_N. This section does not infer the result for its finite central quotients merely from the result for SO_N: the nonlifting spinor classes of those groups require a separate proof.

6 The Half-Spin Finite Lemma in Even Rank D

This section proves the residue-field statement needed for descent in even rank of type D. Throughout, an actual Spin-conjugacy class is fixed. A conjugacy class in the vector representation cannot replace this condition.

6.1 Statement and notation

Let k be an algebraically closed field, let $n = 2m \geq 4$, and assume

$$\operatorname{char} k = 0 \quad \text{or} \quad p \nmid |W(D_n)| = 2^{n-1} n!.$$

Put

$$S = \mathrm{Spin}_{2n}, \qquad \rho : S \longrightarrow \mathrm{SO}_{2n},$$

where ρ is the vector representation, and denote the nonidentity element of $\ker \rho$ by -1. Choose $z \in Z(S)$ such that

$$z^2 = 1, \qquad \rho(z) = -I.$$

There are exactly two such elements z, corresponding to the two half-spin quotients. Let $s \in S(k)$ satisfy

$$\rho(s)^2 = -I, \qquad s^2 = z.$$

Fix the actual $S(k)$-conjugacy class of s.

Proposition 6.1. *Suppose that $x \in S(k)$ satisfies*

$$sxs^{-1} = zx, \qquad \dim Z_S(s,x) = m.$$

Then:

1. *x is regular in S, that is, $\dim Z_S(x) = n$.*
2. *If s' is $S(k)$-conjugate to s and satisfies $s'xs'^{-1} = zx$, then there is a $c \in Z_S(x)(k)$ such that $s' = csc^{-1}$. The full centralizer is used here, retaining all of its connected components.*

The proof proceeds through the spectral decomposition, the Weyl-group action, and the cohomology of the full centralizer.

6.2 Ordinary regularity for every residual spectral type

Proof of Proposition 6.1, part (1). Choose $i \in k$ with $i^2 = -1$. Since $\rho(s)^2 = -I$, the vector space V decomposes as

$$V = V_i \oplus V_{-i}, \qquad \dim V_i = \dim V_{-i} = n.$$

Both subspaces are totally isotropic and are perfectly paired by the quadratic form. In corresponding dual bases, the matrix of the form and the matrix of $\rho(x)$ are

$$J = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \qquad \rho(x) = \begin{pmatrix} 0 & A \\ A^{-T} & 0 \end{pmatrix}, \qquad A \in \mathrm{GL}_n(k).$$

The latter expression follows from $\rho(s)\rho(x) = -\rho(x)\rho(s)$ and preservation of the form. Because n is even, this matrix has determinant one.

The centralizer of s in the vector group is $\mathrm{GL}(V_i)$. Its element $\mathrm{diag}(P, P^{-T})$ acts on A by $A \mapsto PAP^T$. Hence the dimension of the common centralizer is the dimension of the isometry group of the nondegenerate bilinear form A^{-1}. The central Spin covering does not change these dimensions.

Write the Jordan decomposition as $x = x_s x_u$. Since z is central and semisimple,

$$s x_s s^{-1} = z x_s, \qquad s x_u s^{-1} = x_u.$$

Apply the matrix description above to x_s. The asymmetry operator of the bilinear form is

$$D = AA^{-T}.$$

It is also the action of $\rho(x_s)^2$ on V_i, and therefore is semisimple. The bilinear form pairs the λ-eigenspace of D with its λ^{-1}-eigenspace. On the $+1$-eigenspace it is symmetric; on the -1-eigenspace it is alternating.

Write a, b for the dimensions of these two spaces and r_λ for the dimension of each member of each remaining pair $\{\lambda, \lambda^{-1}\}$, where $\lambda \neq \pm 1$. Up to central separable isogeny, the identity component of the common centralizer of the semisimple pair (s, x_s) has factors

$$\mathrm{SO}_a \times \mathrm{Sp}_b \times \prod_{\{\lambda,\lambda^{-1}\}} \mathrm{GL}_{r_\lambda}, \qquad a + b + 2\sum r_\lambda = n.$$

Here b is even, and hence a is even as well. The isometry group of the symmetric form may have additional components, but its unipotent elements belong to SO_a. The sum of the ranks of these factors is exactly

$$\frac{a}{2} + \frac{b}{2} + \sum r_\lambda = m.$$

In each factor the dimension of the centralizer of a unipotent element is at least the rank of that factor. Thus the assumption $\dim Z_S(s, x) = m$ forces x_u to be regular in every factor.

Now consider the ordinary semisimple centralizer $H = Z_S(x_s)$. By the spectral decomposition of x_s^2 above, its factors in the vector representation are

$$\mathrm{SO}_a \times \mathrm{SO}_a \times \mathrm{GL}_b \times \prod_{\{\lambda,\lambda^{-1}\}} (\mathrm{GL}_{r_\lambda} \times \mathrm{GL}_{r_\lambda}).$$

More explicitly, $D = 1$ gives the $+1$ and -1 orthogonal eigenspaces of x_s, while $D = -1$ gives the mutually dual i and $-i$ eigenspaces. Every remaining pair of eigenvalues of the asymmetry operator gives four eigenvalues, and consequently two general linear factors. In these ordinary factors the Jordan partitions of x_u are, respectively,

$$(a-1, 1),\ (a-1, 1); \qquad (b); \qquad (r_\lambda),\ (r_\lambda).$$

These are precisely the regular unipotent partitions in the corresponding factors. If $a = 0$, the corresponding factors are absent. If $a = 2$, the partition is $(1, 1)$, the identity in the torus SO_2, and it remains regular. Thus all degenerate or absent spectral sectors are included.

It follows that x_u is regular in H, and

$$\dim Z_S(x) = \operatorname{rk} H = a + b + 2\sum r_\lambda = n.$$

This proves the first assertion. In particular, the two-block partition $(a - 1, 1)$ in an even-dimensional orthogonal sector is retained throughout. The proof does not assume that the matrix in the vector representation is cyclic. □

6.3 A stable Borel subgroup, a stable torus, and a pinning

Since S is simply connected and semisimple, the centralizer $H = Z_S(x_s)$ of the semisimple element is connected. The preceding subsection proved that x_u is regular in H, so the Borel subgroup B_H of H containing x_u is unique.

The automorphism $\theta = \mathrm{Ad}(s)$ preserves H and fixes x_u; hence it preserves B_H. Moreover, $s^2 = z$ is central, so $\theta^2 = 1$. Since two is invertible, tame first cohomology vanishes for the unipotent groups governing the torsor of maximal tori in the Borel. Consequently B_H contains a θ-stable maximal torus T. Since a semisimple centralizer has the same rank as S, this is also a maximal torus of S. The central element x_s belongs to T.

Let U_H be the unipotent radical of B_H. Regularity means that every simple-root coordinate of x_u in

$$U_H/[U_H, U_H] = \prod_{\alpha \in \Delta_H} U_\alpha$$

is nonzero. Since θ fixes x_u, these coordinates can be used to normalize the simple-root parameters so that θ permutes them without scalar factors. Thus the action of θ on H preserves a pinning, with permutations of isomorphic simple factors allowed. Torus factors have no simple roots and introduce no exception.

This step uses the ordinary regularity already proved. No folding argument is used to assume regularity in the opposite direction.

6.4 The Weyl action and the conjugacy class selected by the exact square

Denote the vector weights of T by $\pm e_1, \dots, \pm e_n$. The cocharacter lattice of Spin is

$$Q = \left\{ (a_1, \dots, a_n) \in \mathbb{Z}^n : \sum a_j \text{ is even} \right\}.$$

Let $w \in W(D_n)$ be the action of s on T. Since $s^2 \in Z(S)$, one has $w^2 = 1$.

A positive one-cycle is impossible. If $w(e_j) = e_j$, then $w(x_s) = z x_s$ and $e_j(z) = -1$ imply

$$e_j(x_s) = -e_j(x_s),$$

contradicting the nonzero eigenvalue and the invertibility of two. A negative one-cycle is also impossible: on the corresponding hyperbolic plane, an isometry interchanging the two isotropic lines has matrix

$$\begin{pmatrix} 0 & c \\ c^{-1} & 0 \end{pmatrix},$$

whose square is I, contrary to $\rho(s)^2 = -I$. Thus w is a product of m signed two-cycles, each with sign product $+1$.

For a pair of coordinates (p, q), choose $\epsilon = \pm 1$ and put

$$\eta = e_p + \epsilon e_q, \qquad \beta = e_p - \epsilon e_q.$$

The root reflection s_η gives the signed interchange on this pair. Its standard root-SL_2 representative n_η satisfies

$$n_\eta^2 = \eta(-1).$$

Taking the product over the disjoint coordinate pairs gives a representative n_w of w. The root subgroups for distinct pairs commute. Changing the sign of one pair changes the corresponding factor in n_w^2 by

$$(2e_q)(-1) = -1 \in \ker \rho.$$

Therefore the two $W(D_n)$-conjugacy classes of these signed two-cycle products are distinguished by the parity of the number of negative pairs. The squares of their standard representatives are z and $-z$, respectively.

It remains to rule out a torus correction changing this square. Write $s = tn_w$. Then

$$s^2 = t\, w(t)\, n_w^2.$$

The image of the norm map $t \mapsto tw(t)$ is the connected fixed torus $(T^w)^0$. Its cocharacter lattice has basis

$$\beta_j = e_{p_j} - \epsilon_j e_{q_j}, \qquad 1 \le j \le m.$$

These vectors span a saturated sublattice of $\mathbb{Z}^n$. Consequently the map from this torus to the vector torus is a closed embedding. In particular, it does not contain the nontrivial kernel element -1 of the Spin covering. Since the vector images of s^2 and n_w^2 are both $-I$, a nonidentity norm correction would have to be precisely that kernel element. The correction is therefore the identity. The exact equality $s^2 = z$ selects one of the two Weyl conjugacy classes above.

The outer automorphism exchanging the two spin nodes also exchanges the two possible elements z. It is therefore enough first to prove the reference case

$$w(e_j) = -e_{n+1-j} \qquad (1 \le j \le n),$$

in which every coordinate pair is interchanged with negative signs. The other half-spin central element is obtained by the outer automorphism.

6.5 The unique nontrivial torus cohomology class

In the reference case, put

$$\begin{aligned} \beta_j &= e_j - e_{n+1-j}, \\ \delta_j &= e_j + e_{n+1-j} \qquad (1 \le j \le m), \\ s_0 &= \prod_{j=1}^{m} n_{\delta_j}. \end{aligned}$$

Then $s_0^2 = z$. A direct calculation gives

$$Q^w = \bigoplus_{j=1}^{m} \mathbb{Z}\beta_j.$$

For $a = (a_1, \dots, a_n) \in Q$,

$$(1+w)a = \sum_{j=1}^{m} (a_j - a_{n+1-j})\beta_j.$$

The sum of these coefficients has the same parity as $\sum a_j$. Thus

$$(1+w)Q = \left\{ \sum b_j \beta_j : \sum b_j \text{ is even} \right\}.$$

Conversely, if $\sum b_j$ is even, choosing $a_j = b_j$ and $a_{n+1-j} = 0$ realizes the required norm, so no values are omitted.

For a torus over an algebraically closed field with a cyclic action of order two invertible in the field, the lattice calculation of cyclic cohomology gives

$$H^1(C_2, T(k)) \simeq Q^w/(1+w)Q \simeq \mathbb{Z}/2.$$

Here we use the usual Tate correspondence between the lattice and the group of torus points. After choosing the root of unity -1, the nonzero class is represented by

$$b = \beta_1(-1).$$

Indeed, $w(\beta_1) = \beta_1$, so $b\, w(b) = 1$, and β_1 is nonzero in the quotient lattice above.

The effect of this class on the actual Spin conjugacy class can be determined explicitly. In the Spin_4 corresponding to the first coordinate pair, β_1, δ_1 are orthogonal roots, and

$$\beta_1(-1)\delta_1(-1) = -1.$$

The root-SL_2 relation also gives

$$\delta_1(i) n_{\delta_1} \delta_1(i)^{-1} = \delta_1(-1) n_{\delta_1}.$$

The root representatives for the other pairs commute with $\delta_1(i)$. Hence

$$\delta_1(i) s_0 \delta_1(i)^{-1} = \delta_1(-1) s_0 = -\beta_1(-1) s_0.$$

Equivalently, $\beta_1(-1)s_0$ is conjugate in S to $-s_0$.

6.6 Half-spin characters exclude the nonzero class

Let χ_+, χ_- be the characters of the two half-spin representations. Each root representative n_{δ_j} is conjugate to $\delta_j(i)$, and these conjugations can be performed simultaneously. Thus the characters of s_0 can be computed on the standard torus.

Choose $r \in k$ with $r^2 = i$. The half-spin weights are $\frac{1}{2}(\pm e_1 \pm \cdots \pm e_n)$, divided into two sets by the parity of the number of minus signs. Therefore

$$\chi_\pm(s_0) = \frac{(r + r^{-1})^{2m} \pm (r - r^{-1})^{2m}}{2}.$$

Since

$$(r + r^{-1})^2 = 2, \qquad (r - r^{-1})^2 = -2,$$

the unordered pair of character values is

$$\{\chi_+(s_0), \chi_-(s_0)\} = \{0, 2^m\}.$$

The kernel element -1 acts as minus the identity in both half-spin representations. Consequently

$$\{\chi_+(-s_0), \chi_-(-s_0)\} = \{0, -2^m\}.$$

Two is invertible in the allowed characteristics, so these unordered pairs differ. This proves that s_0 and $-s_0$ do not belong to the same Spin conjugacy class.

Return now to the actual s in the stable torus. After bringing its Weyl action to the reference w, write $s = t_0 s_0$. The exact square relation gives

$$t_0 w(t_0) = 1.$$

The two cohomology classes computed in the preceding subsection show that, after conjugation by the torus, s is s_0 or $\beta_1(-1)s_0$; the latter is conjugate to $-s_0$. Choose as reference the member of $\{s_0, -s_0\}$ belonging to the actual fixed conjugacy class. The unique nonzero torus class then always sends this reference element to the other Spin conjugacy class.

Thus the condition needed to exclude the nonzero class is that the actual Spin conjugacy class is fixed. Fixing the vector spectrum, or a conjugacy class in the adjoint group, does not suffice.

6.7 The full regular centralizer and uniqueness of conjugacy

Proof of Proposition 6.1, part (2). Put

$$C = Z_S(x) = Z_H(x_u).$$

In the characteristic range under consideration, the standard structure of a regular unipotent centralizer gives

$$C = Z(H) \times U,$$

where U is a connected unipotent group. Here $Z(H)$ must be the full center of multiplicative type, not merely $Z(H)^0$. Both factors are θ-stable.

The unipotent group admits a θ-stable filtration with successive quotients additive groups. On each quotient, cohomology for the order-two action vanishes by averaging with $\frac{1}{2}$. Successive lifting therefore gives

$$H^1(C_2, U(k)) = 1.$$

Thus every first cohomology class of C can be represented by a cocycle in $Z(H)$.

On the other hand, the character lattice of

$$D_H = T/Z(H)$$

is the root lattice of H, with its simple roots as a basis. The pinning constructed in Subsection 6.3 makes θ permute this basis. Hence D_H^θ is a connected torus: in the corresponding permutation coordinates, the fixed points simply require equality of the coordinates on each orbit.

The map

$$T^\theta \longrightarrow D_H^\theta$$

is surjective on k-points. Indeed, the characteristic assumptions of this section make the relevant centers smooth; the prime divisors of their finite orders all occur in $n!$. After passage to Lie algebras, taking invariants under an order-two action is exact, so the differential of this map is surjective. The fixed-point groups themselves are smooth because the action order is invertible. The image is therefore an open subgroup of the connected target torus, and consequently is the entire torus.

The fixed-point and cohomology exact sequence associated with $1 \to Z(H) \to T \to D_H \to 1$ now yields an injection

$$H^1(C_2, Z(H)(k)) \hookrightarrow H^1(C_2, T(k)).$$

Combined with the vanishing of the unipotent part, this embeds $H^1(C_2, C(k))$ into the two torus classes already computed in this section. The map means: first conjugate a cocycle into the full center $Z(H)$, then use $Z(H) \subset T$. It does not treat the entire centralizer C as a subgroup of the torus.

Now take s' as in the proposition and put $a = s's^{-1}$. The two almost-commutation relations imply $a \in C(k)$. Since $s'^2 = s^2 = z$, one also has

$$a\,\theta(a) = 1.$$

Thus a is a first cocycle for the action of C_2 on $C(k)$. Reduce it by a $C(k)$-coboundary to the full center, then map it into torus cohomology. If the resulting torus class were nonzero, Subsection 6.6 would imply that $as = s'$ is conjugate to $-s$, contradicting the assumption that s' is conjugate to the actual s. The class is therefore zero. The injection above then shows that its C-cohomology class is also zero. There is a $c \in C(k)$ such that

$$a = c\,\theta(c)^{-1}, \qquad s' = csc^{-1}.$$

This proves the second assertion. □

6.8 Matching the two actual half-spin endpoints

To connect with the geometric normalization in Section 8, record the actual elements associated with the two maximal endpoints for the same half-spin action. The first endpoint deletes the affine vertices $\{0, n\}$, and its parameter is

$$q = \omega_n = \frac{1}{2}(1, \dots, 1), \qquad s = (2q)(i).$$

Then $\rho(s)^2 = -I$ and $s^2 = z$. The second endpoint deletes $\{1, n-1\}$, with parameter

$$q' = \frac{1}{2}(3, 1, \dots, 1, -1), \qquad s_2 = (2q')(i) = (e_1 - e_n)(-1)s.$$

It has the same exact square z and the same vector spectrum, but is conjugate to $-s$, rather than to s. Explicitly, let v be the even-sign Weyl element that changes the signs of the first and the n-th coordinates simultaneously. Then

$$v(s) = (e_1 + e_n)(-1)s, \qquad (e_1 - e_n)(-1)(e_1 + e_n)(-1) = -1,$$

so $s_2 = -v(s)$.

When Proposition 6.1 is applied at these two endpoints, the actual Spin conjugacy class of s and that of s_2 must therefore be fixed separately. These two classes are exactly interchanged in the half-spin character calculation above, and the proof applies to both. The outer automorphism exchanging the two spin nodes carries the entire argument to the other central element z. This includes both half-spin quotients and both actual endpoints for each quotient.

This section proves the full finite-group lemma. Its application to affine Springer fibers still requires the integral models, smoothness of regular centralizers, and tame descent provided in Section 8.

7 The Primitive-Spinor Finite Lemma in Odd Rank D

7.1 The statement and the actual fourth power

Retain the field and characteristic assumptions of Section 1. Let $n = 2m + 1 \geq 5$, put $S = \mathrm{Spin}_{2n}$, and let $\rho : S \to \mathrm{SO}(V)$ be the vector covering, with $\dim V = 2n$. The center

is $Z(S) \simeq \mu_4$. Let z be either primitive generator, so that $\rho(z) = -I$. Write -1 for the nonidentity element in the kernel of the vector covering.

Fix $s \in S(k)$ whose vector spectrum is

$$1^{2m}, \qquad (-1)^{2m}, \qquad i, \qquad -i, \qquad i^2 = -1.$$

Its actual fourth power is $s^4 = -1$, rather than $s^4 = 1$. We verify this fourth power explicitly below.

Lemma 7.1. *Suppose that $\Gamma \in S(k)$ satisfies*

$$s\Gamma s^{-1} = z^r\Gamma, \qquad r \text{ odd}, \qquad \dim Z_S(s,\Gamma) = m - 1.$$

Then Γ is regular in S in the ordinary, untwisted sense. If s' belongs to the actual S-conjugacy class of s and

$$s'\Gamma(s')^{-1} = z^r\Gamma,$$

then there is an element $b \in Z_S(\Gamma)(k)$ such that $s' = bsb^{-1}$. The centralizer here is the full centralizer, including its finite central components.

7.2 Ordinary regularity for every spectral pattern

Proof of the first assertion of Lemma 7.1. Put $A = \rho(\Gamma)$, and decompose the vector space according to $\rho(s)$:

$$V = V_+ \perp V_- \perp (V_i \oplus V_{-i}).$$

The first two spaces are nondegenerate orthogonal spaces, both of dimension $2m$, while the last two lines are mutually paired isotropic lines. The relation in the vector representation is

$$\rho(s)A = -A\rho(s).$$

Thus A exchanges V_+ and V_-, and also exchanges V_i and V_{-i}.

On the last hyperbolic plane, A is a reflection and has determinant -1. Let the cycle product on the first two orthogonal spaces be

$$B = A^2|_{V_+} \in O(V_+).$$

For a block matrix exchanging two even-dimensional spaces, its determinant is the determinant of this cycle product. Since $\det A = 1$, it follows that $\det B = -1$. Hence $B \in O^-_{2m}(k)$.

An isometry on the first two spaces that commutes with both $\rho(s)$ and A is uniquely determined by an isometry of V_+ commuting with B; the product of the determinants on the two spaces is one. The last hyperbolic plane contributes only a finite centralizer. The separable Spin covering does not change dimensions, so

$$\dim Z_S(s,\Gamma) = \dim Z_{O_{2m}}(B).$$

Let p and q be the dimensions of the $+1$ and -1 eigenspaces of B_s, and let d_a denote the dimension on one side of each of its other reciprocal eigenvalue pairs. Then

$$p + q + 2\sum_a d_a = 2m.$$

Because $\det B = -1$, the integer q is odd. The total dimension is even, so p is odd as well. In particular $p, q \geq 1$. The identity component of the semisimple centralizer is

$$SO_p \times SO_q \times \prod_a GL_{d_a},$$

and its rank is

$$\frac{p-1}{2}+\frac{q-1}{2}+\sum_a d_a = m-1.$$

The minimal dimension of the common centralizer forces B_u to be regular unipotent in each factor. Its natural Jordan blocks are therefore

$$(p), \qquad (q), \qquad (d_a),$$

respectively.

Now return to the block matrix A that exchanges the two spaces. Taking square roots of the $+1$ block of the cycle product produces one block of size p in each of the $+1$ and -1 eigenspaces of A_s. The reflection on the additional hyperbolic plane adds one trivial line to each. The Jordan partition in both spaces is therefore

$$(p,1),$$

the regular unipotent partition for SO_{p+1}. The -1 block of B_s produces a cyclic GL_q block on the reciprocally paired i and $-i$ eigenspaces of A_s. Every other reciprocal pair produces two cyclic GL_{d_a} blocks. Thus the identity factors of the ordinary semisimple centralizer are

$$SO_{p+1}\times SO_{p+1}\times GL_q\times\prod_a (GL_{d_a}\times GL_{d_a}),$$

and the unipotent part is regular in every factor. When $p=1$, the factor SO_2 is a torus, and the trivial unipotent element is regular in that torus. The case $q=1$ is included as well.

Consequently A is regular in the ordinary special orthogonal group. The separable central covering implies that Γ is regular in S. This proves the first assertion without assuming that the residual element is semisimple. □

7.3 A stable pinning from the regular unipotent part

Set $H = Z_S(\Gamma_s)$. A semisimple centralizer in a simply connected semisimple group is connected. By the preceding subsection, Γ_u is regular unipotent in H. The relation with Γ implies that s stabilizes H and fixes Γ_u.

A regular unipotent element belongs to a unique Borel subgroup, so this Borel subgroup is stable under s. Its maximal tori form a torsor under its unipotent radical. Tame vanishing of $H^1(C_4, U)$ supplies a stable maximal torus T. The coordinate of Γ_u in every simple-root direction is nonzero. Normalize the root parameters using these coordinates. Then $\mathrm{Ad}(s)$ permutes the simple-root parameters without scalar factors, giving a stable pinning. This argument also applies when H has torus factors: such factors have no simple-root directions.

7.4 One signed Weyl class for all semisimple sectors

Put $r_0 = (p+1)/2$. The element s exchanges the two SO_{2r_0} factors. Its square acts by $+1$ on the regular block of size p, and by -1 on the added trivial line. Consequently its square is the pinned outer orthogonal involution, which changes the sign of one torus coordinate. The action on the two sets of torus coordinates consists of one negative two-cycle and r_0-1 positive two-cycles. A negative two-cycle means a two-cycle whose product of signs is -1.

When $r_0=1$, the reflection acts on the SO_2 torus by inversion, and the resulting action is still a single negative two-cycle. When $r_0=2$, the outer interchange of the two factors of $D_2 = A_1^2$ has the same description.

On the GL_q factor, the transpose-inverse action induced by s comes from a symmetric form, because $s^2 = 1$ on this sector. After choosing the stable pinning, the torus action is

$$e_j \longmapsto -e_{q+1-j}.$$

Since q is odd, it consists of one negative one-cycle and $(q-1)/2$ positive two-cycles. Every exchanged pair of GL_{d_a} factors contributes d_a positive two-cycles. The total number of positive two-cycles is therefore

$$\frac{p-1}{2} + \frac{q-1}{2} + \sum_a d_a = m - 1.$$

Conjugating by signed coordinate changes puts each positive two-cycle into the form of an exchange with two minus signs. If the conjugating change uses an odd number of sign changes, make one more sign change on the coordinate of the negative one-cycle. This does not change the action, and it makes the total number of sign changes even. Thus the conjugacy takes place inside $W(D_n)$, rather than merely inside the larger signed symmetric group. We may take the standard form

$$\begin{aligned} w(e_1) &= -e_n, & w(e_n) &= e_1, \\ w(e_j) &= -e_{n+1-j} & & (2 \le j \le n-1). \end{aligned}$$

The first pair is a negative two-cycle, the middle coordinate is a negative one-cycle, and all remaining pairs are positive two-cycles.

7.5 The actual fourth power of a reference lift

Choose a vector Weyl representative on a hyperbolic basis, and denote one of its Spin lifts by s_0. A positive two-cycle has spectrum $1, 1, -1, -1$, a negative one-cycle has spectrum $1, -1$, and a negative two-cycle has spectrum $1, -1, i, -i$. Its total spectrum is therefore exactly

$$1^{2m}, \qquad (-1)^{2m}, \qquad i, \qquad -i.$$

The vector element $\rho(s_0)$ is conjugate to $\rho(s)$ in $SO(V)$. Indeed, if an initial conjugating isometry has determinant -1, multiplying it by a reflection in a nonzero $+1$ or -1 eigenspace changes its determinant to $+1$.

This vector element is conjugate to $q(i)$, where

$$q = 2(e_1 + \cdots + e_m) + e_{m+1}.$$

The vector q is an SO cocharacter, but its coordinate sum is odd, so it is not a Spin cocharacter. A Spin lift can be written as

$$(2q)(\zeta_8), \qquad \zeta_8^2 = i.$$

Its fourth power is $(2q)(-1) = -1$. Thus the actual element satisfies $s^4 = -1$, and every Spin lift of the chosen vector representative has the same fourth power. The two lifts differ by the vector kernel element -1, which does not change their fourth powers.

After identifying the stable torus, write

$$s = as_0, \qquad a \in T(k).$$

Equality of the actual fourth powers gives

$$a\, w(a)\, w^2(a)\, w^3(a) = 1.$$

Thus a is a genuine torus cocycle, rather than a solution to a fourth-power equation with an unremoved central twist.

7.6 All cohomology classes for the order-four action

The Spin cocharacter lattice is

$$Q(D_n) = \left\{ v \in \mathbb{Z}^n : \sum_j v_j \text{ is even} \right\}.$$

The invariant lattice for the standard w has basis

$$\beta_j = e_j - e_{n+1-j}, \qquad 2 \le j \le m.$$

Let $N = 1 + w + w^2 + w^3$. On a positive two-cycle, a norm coefficient is twice the coordinate difference. The coordinate of the negative one-cycle can change the parity of the total coordinate sum without changing the norm. The negative two-cycle also has zero norm. It follows that

$$NQ(D_n) = 2Q(D_n)^w.$$

The tame cyclic torus cohomology formula gives

$$H^1(C_4, T(k)) \simeq Q(D_n)^w / NQ(D_n) \simeq (\mathbb{Z}/2)^{m-1}.$$

Actual point representatives for these classes are

$$c_J = \prod_{j \in J} \beta_j(i), \qquad J \subset \{2, \dots, m\}.$$

Here i cannot be replaced by -1: for an order-four action, the representative $\beta_j(-1)$ corresponds to twice the lattice class, which is already zero. Multiplication by $\beta_j(i)$ changes the square on the four-dimensional vector block associated to that positive two-cycle from $+1$ to -1. Hence

$$\dim \ker\bigl(\rho(c_J s_0)^2 + I\bigr) = 2 + 4|J|.$$

For the actual s, this dimension is two. Therefore no nonempty J gives an element conjugate to s.

First apply this exclusion to a. Its cohomology class is zero, so after T-conjugation we may arrange that the actual $s = s_0$. The exclusion consequently applies to the actual action $\mathrm{Ad}(s)$, and not just to a formal reference representative.

7.7 Injecting cohomology of the full centralizer

Since Γ_u is regular in H, its full centralizer is

$$C_\Gamma = Z_S(\Gamma) = Z_H(\Gamma_u) = Z(H) \times U_C,$$

where U_C is connected unipotent and $Z(H)$ includes all finite components. Both factors are stable under s. Tame vanishing for unipotent cohomology yields

$$H^1(C_4, C_\Gamma(k)) \simeq H^1(C_4, Z(H)(k)).$$

The character lattice of the torus $Q_T = T/Z(H)$ is the root lattice of H, and its simple-root basis is permuted by s. Thus Q_T^s is connected: the coinvariants of a permutation lattice are torsion-free. Our characteristic assumptions exclude the relevant finite orders in the center, so $Z(H)$ is smooth. Taking invariants for an action of invertible order preserves exactness in

$$0 \longrightarrow \mathrm{Lie}\, Z(H) \longrightarrow \mathrm{Lie}\, T \longrightarrow \mathrm{Lie}\, Q_T \longrightarrow 0.$$

Consequently the map $T^s \to Q_T^s$ has surjective differential. Its image is a closed subgroup of full dimension in a connected target, and hence the map is surjective on k-points. The cohomology exact sequence gives an injection

$$H^1(C_4, C_\Gamma(k)) \simeq H^1(C_4, Z(H)(k)) \hookrightarrow H^1(C_4, T(k)).$$

This arrow does not embed the group C_Γ, with its unipotent factor, into a torus. It first removes the unipotent cohomology and then uses the inclusion of the full center into the torus.

7.8 Full-centralizer conjugacy of competing inertia elements

Proof of the second assertion of Lemma 7.1. Let s' satisfy the second assertion of the lemma. Then

$$d = s's^{-1} \in C_\Gamma(k),$$

and

$$d\ \mathrm{Ad}(s)(d)\ \mathrm{Ad}(s)^2(d)\ \mathrm{Ad}(s)^3(d) = 1.$$

The cocycle equation follows from $(s')^4 = s^4 = -1$.

First conjugate s' by an element of $U_C(k)$ to remove its unipotent cocycle component, and then map the remaining central cocycle to torus cohomology. Its class has a representative c_J. Further torus conjugation therefore makes the resulting inertia element equal to $c_J s$.

If $J \neq \varnothing$, the dimension $2 + 4|J|$ determined by its square spectrum contradicts the dimension two for the actual conjugacy class. Thus its torus class is zero. By the injection above, its full-centralizer cohomology class is also zero. This is exactly the assertion that

$$s' = bsb^{-1}, \qquad b \in C_\Gamma(k).$$

The second assertion is proved. The argument applies uniformly to every $m \geq 2$, every allowed residual semisimple sector, and nonsemisimple regular reductions. □

8 Uniform Tame Normalization and Integral Descent

This section turns the finite almost-commuting lemmas in Sections 6, 7, and 9 into regular-point transitivity over the original field L. The actual simply connected inertia lift, the full residue centralizer, and every Kottwitz grade are retained.

8.1 The cyclic central envelope and its parameters

Let S be split simply connected semisimple and let $C = \langle z \rangle \subset Z(S)$ be cyclic of order e. Choose $\iota : C \xrightarrow{\sim} \mu_e$, and put

$$H_C = (S \times \mathbb{G}_m)/\{(c, \iota(c)^{-1}) : c \in C\}, \qquad \chi([h, a]) = a^e.$$

The derived group embeds as S, and $H_C/S = \mathbb{G}_m$, so

$$\pi_1(H_C) = \mathbb{Z}, \qquad \kappa_{H_C} = v_L \circ \chi.$$

The scalar central cocharacter has grade e. For $S = \mathrm{Spin}_{4m}$ with a half-spin μ_2 glued, the other order-two center survives; the full center is not declared connected.

The following parameters use the standard orthogonal or Bourbaki coordinates. The point x is the facet barycenter and $q = ex$ is an adjoint cocharacter.

Type of S	e	Deleted vertices	q	Minimum dimension
D_{2m+1}, $m \geq 2$	4	$\{m, m+1\}$	$2\sum_{j=1}^{m} e_j + e_{m+1}$	$m-1$
D_{2m}, $m \geq 2$	2	$\{0, 2m\}$	$\omega_{2m} = (1, \ldots, 1)/2$	m
Second even-D facet	2	$\{1, 2m-1\}$	$(3, 1, \ldots, 1, -1)/2$	m
E_6	3	$\{0, 1, 6\}$	$\omega_1 + \omega_6$	2
E_7	2	$\{0, 7\}$	ω_7	4

The other even-D half-spin action exchanges the spinor ends. The two odd-D primitive classes are inverse and have the same facet; the finite lemma covers both odd exponents.

8.2 All maximizing spinorial facets

Write n for the rank of D_n. Deleting an internal orbit $\{j, n-j\}$ leaves

$$D_j \times A_{n-2j-1} \times D_j,$$

with log-volume

$$f(j) = 2j(j-1) + \frac{(n-2j-1)(n-2j)}{2}.$$

This is strictly convex. For $n = 2m+1$, the four-endpoint deletion has volume $(m-1)(2m-1)$, while $2 \leq j \leq m$ gives

$$f(m) = 2m(m-1), \qquad f(m) - f(2) = 5(m-2),$$

$$f(m) - (m-1)(2m-1) = m - 1 > 0.$$

For $m = 2$ the middle pair is the sole internal orbit. For $m > 2$, convexity and these differences make $\{m, m+1\}$ the unique maximum.

For $n = 2m$, either endpoint-pair deletion leaves A_{n-1}, of volume

$$a = \frac{n(n-1)}{2} = m(2m-1).$$

When the internal range $2 \leq j \leq m-1$ is nonempty,

$$a - f(2) = 4n - 14 > 0, \qquad a - f(m-1) = 5(m-1) > 0.$$

Deleting the fixed middle vertex leaves $D_m \times D_m$, of volume $2m(m-1) = a - m < a$. Exactly the two displayed facets maximize. Deleting several orbits removes further roots from a single-orbit deletion and strictly reduces the positive-root count. The boundary ranks D_4, D_5 are included; D_2, D_3 are type A.

8.3 Actual inertia lifts and semilinear descent

Take

$$F = L(u), \qquad u^e = t, \qquad \sigma(u) = \zeta u,$$

with ζ primitive of order e and $\iota(z) = \zeta$. Strict henselian completeness and invertibility of e supply the roots of unity and roots of units, also in mixed characteristic. Even when q is not a cocharacter of S,

$$A = \mathrm{Ad}(q(u))$$

is an F-defined automorphism of S. Choose $s \in S(\mathcal{O})$ lifting $q(\zeta)^{-1}$, and set

$$\delta = \mathrm{Ad}(s) \circ \sigma.$$

Since $\sigma(q(u)) = q(\zeta)q(u)$ and s^e is central,

$$\delta^e = 1, \qquad \delta(A(h)) = A(\sigma(h)).$$

The actual lifts can be chosen as follows.

- In odd D, choose $\eta^2 = \zeta$ and take $s = (2q)(\eta^{-1})$. Here $2q$ lies in the Spin coroot lattice but q does not, and $s^4 = -1$, the vector-cover kernel. Its vector spectrum is $1^{2m}, (-1)^{2m}, i, -i$.
- In even D, take $s = (2q)(i)$, $i^2 = -1$. Then $\rho(s)^2 = -I$ and $s^2 = z$. At the second facet

  $$s_2 = (e_1 - e_n)(-1)s$$

  is Spin-conjugate to $-s$, not s; each actual class is fixed separately.
- In E_6, take $s = (\omega_1 + \omega_6)(\zeta^{-1})$, so $s^3 = 1$.
- In E_7, take $s = (2\omega_7)(i)$, so $s^2 = z$.

The required order is that of $\mathrm{Ad}(s)$, not of s. Quadratic descent permits an order-four lift, and odd-D quartic descent permits an order-eight lift.

8.4 The fixed model of the connected parahoric

For a root parameter $a \in L$,

$$A(x_\alpha(a)) = x_\alpha(u^{\alpha(q)}a), \qquad v_F(u^{\alpha(q)}a) = e\, v_L(a) + \alpha(q).$$

Thus integral fixed elements have the original root bound

$$v_L(a) \geq \lceil -\alpha(x) \rceil.$$

Consider

$$\mathscr{P} = \left(\mathrm{Res}_{\mathcal{O}_F/\mathcal{O}}\, S_{\mathcal{O}_F}\right)^\delta.$$

Tame fixed-point smoothness makes this model smooth. Its generic fiber is S_L through A, and its root bounds and torus model agree with the connected parahoric.

Reduction modulo u maps onto $Z_{S_k}(s)$. The kernel has a jet filtration with vector-group quotients; tame invariants preserve connected unipotence. The split integral centralizer supplies constant lifts. Since S is simply connected, $Z_{S_k}(s)$ is connected. This identifies the connected model, not the full facet normalizer. As $\Omega_S = 1$, the facet stabilizer in $S(L)$ is P_x, and

$$A(P_x) = S(\mathcal{O}_F)^\delta.$$

The odd-D facet does not contain vertex 0; root bounds, not a finite-parabolic preimage inside $S(\mathcal{O})$, are used.

The surviving roots satisfy $\alpha(q) \equiv 0 \pmod{e}$, giving

S	derived type of $Z_S(s)$	central rank
D_{2m+1}	$D_m \times D_m$	1
D_{2m}	A_{2m-1}	1
E_6	D_4	2
E_7	E_6	1

with their full torus lattices. The H_C model is $(\mathscr{P} \times \mathbb{G}_m)/C$, with reductive quotient $(Z_S(s) \times \mathbb{G}_m)/C$. The finite quotient is separable and $H^1(\mathcal{O}, C) = 0$ by unit-root extraction. The full facet normalizer modulo the connected parahoric is $\mathbb{Z}$, measured by $v(\chi)$; only its diagram action factors through $\mathbb{Z}/e$.

8.5 Central normalization of the fixed element

Let $\gamma \in P_x^C \dot{\tau}^r$, with $\gcd(r, e) = 1$ in the primitive cases. Write

$$\chi(\gamma) = t^r a, \quad a \in \mathcal{O}^\times, \qquad \eta_0^e = a, \qquad c = u^r \eta_0.$$

Over F, write uniquely $\gamma = [h, c]$, $h \in S(F)$. Galois invariance gives

$$\sigma(h) = z^{-r} h.$$

Because γ fixes x, scalar normalization leaves the derived building unchanged, and A moves that point to the hyperspecial origin. Hence $X = A(h) \in S(\mathcal{O}_F)$ and

$$\boxed{\delta(X) = z^{-r} X.}$$

Only the residue $\Gamma = \overline{X}$ satisfies the ordinary relation

$$s\Gamma s^{-1} = z^{-r}\Gamma.$$

The integral semilinear equation cannot be replaced by ordinary conjugation.

8.6 The actual reductive-coset twist

The normalized residue N of a length-zero representative gives $\Gamma \in Z_S(s)N$, with the following actual actions.

- Odd D: exchange of the D_m factors, whose square is the orthogonal outer involution, and inversion on the extra central torus; fixed semisimple rank $m - 1$.
- Even D: transpose-inverse on A_{2m-1} and inversion on the center; fixed rank m.
- E_6: triality on D_4, with no fixed direction on the extra rank-two center; fixed rank 2.
- E_7: the diagram involution on E_6, with inversion on the extra center; fixed rank 4.

The classical actions follow from the vector blocks; the exceptional ones are checked from affine vertex permutations and central lattices. The scalar factor adds one to both actual and minimum centralizer dimensions. Original coset regularity is therefore exactly

$$\dim Z_S(s, \Gamma) = \begin{cases} m - 1, & D_{2m+1}, \\ m, & D_{2m}, \\ 2, & E_6, \\ 4, & E_7. \end{cases}$$

These are the finite-lemma hypotheses, not ordinary regularity assumed beforehand.

8.7 An integral Steinberg transporter

Lemma 8.1. *Let R be a complete strictly henselian DVR in the permitted characteristic, let $E = \operatorname{Frac} R$, and let S_R be split simply connected. If $X_1, X_2 \in S(R)$ are $S(E)$-conjugate and have ordinary regular residues, then an $a \in S(R)$ satisfies $aX_1a^{-1} = X_2$. The full scheme $Z_{S_R}(X_1)$ is smooth.*

Proof. Use the integral fundamental-character coordinates of the quotient

$$\chi_S : S_R \longrightarrow \mathbb{A}_R^{\operatorname{rk} S}.$$

The field-level Steinberg theorem on both geometric fibers gives smoothness on the regular locus, a single orbit per regular fiber, and smooth regular centralizers. A full-rank quotient

Jacobian minor is a unit at a regular residue, so the integral quotient is smooth near the required points. For a geometric regular X,

$$\ker(d\chi_{S,X}) = T_X(\text{conjugacy orbit});$$

both have dimension $\dim S - \operatorname{rk} S$. Consequently

$$S_R \times S_{\mathrm{reg}} \longrightarrow S_{\mathrm{reg}} \times_{\mathbb{A}_R^{\operatorname{rk} S}} S_{\mathrm{reg}}, \qquad (a, X) \longmapsto (aXa^{-1}, X)$$

has surjective differential and is smooth, since its source and target are smooth on these neighborhoods. Generic conjugacy gives equality of quotient coordinates in E, hence in R. The transporter is the resulting smooth pullback; the single-orbit theorem supplies a residue-field point, and strict henselian completeness lifts it. Pullback along (X_1, X_1) gives the smooth full centralizer. □

The lemma is applied to split $S_{\mathcal{O}_F}$, without invoking an affine Springer transitivity theorem for a ramified group.

8.8 Grade-zero representatives of two regular flags

Translate two regular H_C flag points to grade zero using Lemma 3.1, which covers the full $\pi_1(H_C) = \mathbb{Z}$. Their χ-values are then units. Extracting scalar unit roots and absorbing them into the parahoric gives representatives

$$h_1, h_2 \in S(L).$$

Put $X_i = A(h_i)^{-1} X A(h_i)$. They are integral, generically conjugate, and satisfy $\delta(X_i) = z^{-r} X_i$. The first conclusion of the finite lemmas makes their residues ordinary regular. Lemma 8.1 gives

$$a \in S(\mathcal{O}_F), \qquad aX_1a^{-1} = X_2.$$

8.9 Killing the full residue-centralizer cocycle

Let $\mathscr{J} = Z_{S_{\mathcal{O}_F}}(X_1)$. It is smooth and δ-stable, since $\delta(X_1)$ is a central multiple of X_1. The discrepancy

$$d = a^{-1}\delta(a) \in \mathscr{J}(\mathcal{O}_F)$$

is a genuine cyclic cocycle. Reducing modulo u,

$$\overline{d} = s's^{-1}, \qquad s' = \overline{a}^{-1} s \overline{a}.$$

The element s' is conjugate to the actual s in S and obeys the same central relation for $\overline{X}_1$. The second finite conclusion provides

$$\overline{b} \in Z_S(\overline{X}_1)(k) = \mathscr{J}(k), \qquad s' = \overline{b}\, s\, \overline{b}^{-1}.$$

Hence $\overline{d} = \overline{b}\,\delta(\overline{b})^{-1}$. Lift $\overline{b}$ by smoothness and replace a by ab; the new cocycle is trivial modulo u. The full residue centralizer is essential: its finite central components can supply the first correction.

8.10 Congruence averaging and descent to the original field

For

$$\mathscr{J}^{\ell} = \ker\big(\mathscr{J}(\mathcal{O}_F) \to \mathscr{J}(\mathcal{O}_F/u^{\ell})\big), \qquad \ell \geq 1,$$

smoothness gives vector-group quotients

$$\mathscr{J}^{\ell}/\mathscr{J}^{\ell+1} \simeq \mathrm{Lie}(\mathscr{J}_k) \otimes_k u^{\ell}/u^{\ell+1}.$$

The cyclic action includes ζ^{ℓ}. Averaging by $1/e$ gives $H^1(C_e, \mathscr{J}^{\ell}/\mathscr{J}^{\ell+1}) = 0$. If the discrepancy lies in $\mathscr{J}^{\ell}$, its next graded class is an additive cocycle and can be killed by a lifted coboundary. Induction kills it to every order.

The correcting product converges by completeness to

$$a_\infty X_1 a_\infty^{-1} = X_2, \qquad \delta(a_\infty) = a_\infty, \qquad a_\infty \in S(\mathcal{O}_F).$$

The identity $\delta A = A\sigma$ makes $a_\infty = A(a_L)$ with $a_L \in S(L)$, and root bounds put $a_L \in P_x$. Thus $h_2 a_L h_1^{-1}$ centralizes γ and sends $h_1 P_x^C$ to $h_2 P_x^C$. Because H_C has simply connected derived group, its generic regular-semisimple centralizer is a connected torus. Restoring the initial grade translations proves $\mathsf{L}(H_C)$. The final element lies in the connected parahoric with grade zero; the initial translations used the full integer grades, so no normalizer decoration is lost.

8.11 The identity-action class and adjoint spinorial classes

For a grade divisible by e, a scalar central cocharacter removes that grade, and a scalar unit root gives the ordinary hyperspecial problem in S. He's Theorem 3.5 applies. Different hyperspecial types may be transported by the L-group automorphism defined by the relevant adjoint point; that point need not lift to $S(L)$.

For a D_n spinorial class a, choose the corresponding $C = \langle z \rangle$. First descend through the split scalar kernel

$$H_C \longrightarrow S/C.$$

Then transfer through

$$S/C \longrightarrow S/Z(S),$$

whose Kottwitz image is $\langle a \rangle$. The fixed target element of grade a does lift. Odd primitive D uses $C = \mu_4$ and already reaches the adjoint group; even D uses each actual half-spin subgroup, not the vector SO kernel. The vector class comes from Section 5 and the zero class from ordinary Spin. Thus every adjoint D class is included. For E_6, E_7, take $C = Z(S)$; the finite lemmas in Section 9 supply all nonzero adjoint classes by the same split-scalar descent.

9 Exceptional Types

This section proves all the local assertions for exceptional types and then applies them to every Newton Levi subgroup. Write S for a split, simply connected simple group and S_{ad} for its adjoint group. Throughout, we impose the field and characteristic assumptions of Section 1, in particular $\mathrm{char}\, k = 0$ or $p \nmid |W(G)|$. These assumptions remain valid for every Levi subgroup and every simple factor of its root system.

9.1 Covered cases and the remaining local assertions

The algebraic fundamental groups of the simple adjoint exceptional groups are as follows:

Type	G_2	F_4	E_6	E_7	E_8
$\pi_1(S_{\mathrm{ad}})$	0	0	$\mathbb{Z}/3$	$\mathbb{Z}/2$	0.

(1)

In the basic adjoint case, the Newton vector is zero. Thus G_2, F_4, E_8 have only the zero Kottwitz class, their length-zero action is the identity, and their endpoints of maximal volume are hyperspecial. He's theorem for the ordinary integral case, recalled in Section 2, gives the required local transitivity. The centralizer of a regular semisimple element in any of these groups is a connected torus, so the acting group here is precisely the identity component of the centralizer.

The zero adjoint Kottwitz classes in E_6, E_7 lift to the simply connected groups and are also handled by the ordinary integral case and the fixed-element lifting lemma of Section 3. The remaining classes are exactly the two nonzero classes in E_6 and the unique nonzero class in E_7. They are handled by the finite-dimensional lemmas below and the tame integral descent lemma of Section 8. We do not identify these nonzero Kottwitz classes with the ordinary integral case, even though their Newton vector is zero.

9.2 All very special endpoints for nonzero classes and the exact inertia lifts

We use Bourbaki numbering: the main chain in E_6, E_7, E_8 is $1-3-4-5-6(-7-8)$, and node 2 is attached to node 4. Write $\omega_i^\vee$ for the fundamental coweights. Roots and coroots are naturally identified in these simply laced root systems, but all quantities used below to act on tori are coweights or coroots.

Let z generate $Z(S)$, and put $e = |Z(S)|$. Choose a primitive e-th root of unity $\zeta \in \mathcal{O}$, and fix an isomorphism $\iota : Z(S) \longrightarrow \mu_e$ with $\iota(z) = \zeta$. Construct the central envelope

$$H^+ = (S \times \mathbb{G}_m)/\{(c, \iota(c)^{-1}) : c \in Z(S)\}, \qquad \chi([h,a]) = a^e. \tag{2}$$

Its derived group is S, its algebraic fundamental group is $\pi_1(H^+) = \mathbb{Z}$, and the Kottwitz grade is measured by $v(\chi)$. The grade of a scalar central cocharacter is e.

The node orbits of the nonzero length-zero actions and the volumes obtained by deleting one orbit are listed below. Here volume comparisons mean comparisons of the numbers of positive roots in the reductive quotients.

Type	Node orbits of the nonzero action	Volumes after deleting the respective orbits
E_6	$\{0,1,6\},\ \{2,3,5\},\ \{4\}$	12, 4, 9
E_7	$\{0,7\},\ \{1,6\},\ \{3,5\},\ \{2\},\ \{4\}$	36, 14, 9, 28, 13

Deleting at least two orbits removes further roots and strictly decreases the volume. Consequently, the unique very special endpoint for a nonzero E_6 class deletes $\{0,1,6\}$, and the unique very special endpoint for the nonzero E_7 class deletes $\{0,7\}$. The inverse generator in E_6 has the same node orbits and the same maximal endpoint.

The corresponding fixed points, cocharacters, and actual simply connected inertia lifts are the following.

Type	e	x	$q = ex$	Actual lift s	s^e
E_6	3	$(\omega_1^\vee + \omega_6^\vee)/3$	$\omega_1^\vee + \omega_6^\vee$	$q(\zeta^{-1})$	1
E_7	2	$\omega_7^\vee/2$	$\omega_7^\vee$	$(2\omega_7^\vee)(i)$, $i^2 = -1$	z

For E_6, the explicit integral coroot expression is

$$q = 2\alpha_1^\vee + 2\alpha_2^\vee + 3\alpha_3^\vee + 4\alpha_4^\vee + 3\alpha_5^\vee + 2\alpha_6^\vee. \tag{3}$$

Thus q itself is a cocharacter of S. For E_7, q is an adjoint cocharacter and $2q$ belongs to the coroot lattice. The action of q in the adjoint group still defines an inner automorphism of S. In both cases, the adjoint image of s is $q(\zeta)^{-1}$. In particular, the actual E_7 lift s has order four although its adjoint image has order two. The cocycle conditions below retain $s^2 = z$; they do not replace it with a lift having square 1.

A root subgroup U_α survives in the reductive quotient if and only if e divides $\alpha(q)$. For E_6, the sum of the coefficients at nodes $1, 6$ of a positive root is $0, 1$, or 2; the surviving roots are therefore exactly a D_4 root system. For E_7, the coefficient at node 7 of a positive root is 0 or 1; the surviving roots are exactly an E_6 root system. The torus ranks remain $6, 7$, respectively. Hence

$$\begin{gathered}\text{The derived types and central ranks of } Z_S(s) \text{ are } (D_4, 2) \text{ and } (E_6, 1),\\ \dim Z_S(s) = 30,\ 79,\end{gathered} \tag{4}$$

respectively.

The normalized residual length-zero action is triality on D_4 and diagram reversal on E_6. Its fixed rank on the derived torus is 2 in the former case and 4 in the latter. The fixed ranks of the full affine Weyl actions are also $2, 4$, respectively, so the additional two-dimensional and one-dimensional central directions have no nonzero fixed vectors. The minimal centralizer dimensions in these actual disconnected cosets are therefore $2, 4$. Incorporating the scalar center in (2) increases both dimensions by one.

9.3 A common statement of the two finite-dimensional lemmas

Lemma 9.1 (Finite almost-commuting lemma)**.** *Use the actual s of Subsection 9.2. Suppose that $\Gamma \in S(k)$ satisfies*

$$s\Gamma s^{-1} = z^{-r}\Gamma, \qquad e \nmid r, \qquad \dim Z_S(s, \Gamma) = \begin{cases} 2, & S \text{ has type } E_6,\\ 4, & S \text{ has type } E_7. \end{cases} \tag{5}$$

Then Γ is an ordinary regular element of S. Moreover, if s' is conjugate to the actual s in $S(k)$ and satisfies the same almost-commuting relation with Γ, there exists $c \in Z_S(\Gamma)(k)$, in the full centralizer, such that $s' = csc^{-1}$.

We prove both assertions below. The proof includes every semisimple Jordan sector and every unipotent residual pattern.

9.4 Choosing a stable Borel subgroup and torus first

Write $\Gamma = \Gamma_s\Gamma_u$, and let $R = Z_S(\Gamma_s)$. Since S is simply connected, the semisimple centralizer R is connected and reductive and has the full rank of S. Relation (5) shows that $\theta = \mathrm{Ad}(s)$ preserves R, with $\theta(\Gamma_s) = z^{-r}\Gamma_s$ and $\theta(\Gamma_u) = \Gamma_u$.

We use the stable Borel theorem for finite-order automorphisms: over an algebraically closed field, an automorphism of a connected reductive group whose finite order is prime to

the characteristic stabilizes a Borel subgroup and a maximal torus in it. Equivalently, it is a quasi-semisimple element in the corresponding disconnected reductive semidirect product. Here the order of θ divides e, so we first obtain a θ-stable pair $T \subset B \subset R$, without using any conclusion about the regularity of Γ_u.

The torus T is also maximal in S, and $\Gamma_s \in Z(R) \subset T$. Let w be the Weyl element induced by s on T. For each w-invariant weight $\lambda \in X^*(T)$, we have

$$\lambda(\Gamma_s) = \lambda(w(\Gamma_s)) = \lambda(z^{-r})\lambda(\Gamma_s), \qquad \lambda(z^{-r}) = 1. \tag{6}$$

This central-charge condition excludes the inadmissible Weyl classes.

9.5 The Weyl classes in E_6 and all fixed-root phases

We use the finite Carter-class classification of $W(E_6)$: the nonidentity order-three classes are exactly $A_2, 2A_2, 3A_2$. The complete data required here are as follows.

Carter class	Fixed Cartan rank	Number of fixed roots	Class size
A_2	4	12	240
$2A_2$	2	6	480
$3A_2$	0	0	80

This table can also be obtained by finite integral-lattice operations: generate all 51840 integral Weyl matrices using

$$s_i(\alpha_j) = \alpha_j - \langle \alpha_j, \alpha_i^\vee \rangle \alpha_i,$$

select the order-three matrices, and take conjugacy closures under the six generators. This gives the three classes in the table. We are using the complete finite root-system classification, rather than deducing geometric transitivity from numerical checks in finitely many orders.

A representative of the A_2 class is $s_1 s_3$. It fixes the fundamental weight ω_6, which is nontrivial on $Z(S) = \mu_3$; the relevant inverse-Cartan entry is $2/3$. This class is therefore excluded by (6). The identity class is excluded in the same way.

If w belongs to $3A_2$, there are no fixed Cartan directions and the 72 roots form 24 three-cycles. The product of the phases around each three-cycle is 1, so each contributes one fixed direction. This would give $\dim Z_S(s) = 24$, contradicting (4). Thus w must belong to $2A_2$, and it may be conjugated to the linear part of the affine rotation

$$(0\ 1\ 6)(2\ 3\ 5), \qquad 4 \text{ fixed}, \tag{7}$$

or to its inverse. Its fixed Cartan dimension is 2. Its six fixed roots form A_2, and the other 66 roots form 22 three-cycles. If f_+ is the number of fixed-root spaces with phase 1, then

$$\dim Z_S(s) = 2 + 22 + f_+ = 30. \tag{8}$$

Consequently $f_+ = 6$: the phase on every fixed-root space is 1. This compares integer eigenspace dimensions; it does not reduce equality of traces modulo the characteristic. Replacing ζ by ζ^{-1}, or replacing r by the other nonzero class modulo 3, merely interchanges the nontrivial phases. The argument thus handles both nonzero E_6 classes.

9.6 Orthogonal-reflection classification in E_7 and fixed-root phases

Every involution in the Weyl group is a product of reflections in mutually orthogonal roots. Let a be the number of reflections, equivalently the dimension of the negative Cartan eigenspace.

If $a = 0$, a weight with nontrivial central charge is automatically fixed, contradicting (6). For $a = 1$ or 2, transitivity on roots and on orthogonal root pairs allows us to use s_1 or $s_1 s_2$ as a representative. Both fix ω_7, whereas $\omega_7(z) = -1$; its central pairing is $3/2$. These cases are likewise inadmissible.

If $a \geq 4$, the fixed Cartan rank is $r_0 = 7 - a \leq 3$. The fixed root system has rank at most 3, so its number f of roots is at most 12. Each nonfixed pair of roots contributes one fixed direction. Hence

$$\dim \mathfrak{s}^{\mathrm{Ad}\, s} \leq r_0 + \frac{126 - f}{2} + f \leq 3 + \frac{126 + 12}{2} = 72 < 79. \tag{9}$$

Only $a = 3$ remains. Its classification can be proved directly without enumerating $W(E_7)$. The perpendicular root system of one root is D_6; the common perpendicular root system of two orthogonal roots is $D_4 + A_1$. A third perpendicular root lies in either the D_4 or the A_1 factor, and the Weyl group stabilizing the first two roots acts transitively on each kind of choice. The resulting two classes have common perpendicular root systems A_1^4 and D_4, with 8 and 24 fixed roots, respectively.

For the first class, the fixed dimension is at most $4 + (126 + 8)/2 = 71$, so it is inadmissible. The second class has fixed Cartan rank 4, and its nonfixed roots form 51 pairs. Therefore

$$79 = \dim \mathfrak{s}^{\mathrm{Ad}\, s} = 4 + 51 + f_+, \qquad f_+ = 24. \tag{10}$$

Every fixed-root space has phase $+1$. Again, this is an integer dimension calculation and is unchanged in the allowed positive characteristics.

We may take w to be the linear part of the affine Ω-involution:

$$\alpha_1 \leftrightarrow \alpha_6, \quad \alpha_3 \leftrightarrow \alpha_5, \quad \alpha_2 \mapsto \alpha_2, \quad \alpha_4 \mapsto \alpha_4, \quad \alpha_7 \mapsto -\widetilde{\alpha}, \tag{11}$$

where

$$\widetilde{\alpha} = 2\alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + \alpha_7. \tag{12}$$

A simple system of the fixed D_4 consists of the central root α_2 and the three arms

$$\alpha_4, \qquad \alpha_3 + \alpha_4 + \alpha_5, \qquad \alpha_1 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6. \tag{13}$$

9.7 From folded regularity to ordinary regularity for every Jordan pattern

In the stable Borel pair chosen in Subsection 9.4, θ permutes the simple roots of R. We have proved that the phase on every fixed simple-root space is 1; the product of the phases on each length-e orbit of simple-root spaces is also 1. Rescaling root parameters successively around each orbit makes θ preserve a pinning. Its action on the central torus is retained.

All roots within any simple-root orbit are mutually orthogonal. For an order-three action, the only possibilities on the diagram are a cyclic permutation of isomorphic factors and D_4 triality, for which the assertion holds. For an order-two action, suppose an orbit contained adjacent simple roots α, β. After normalizing the root vectors, the phase of θ on $\mathfrak{g}_{\alpha+\beta} = k[X_\alpha, X_\beta]$ would be -1, since θ interchanges the two entries of the bracket. But $\alpha + \beta$ is a w-fixed root of the ambient system, and Subsection 9.6 shows that its phase must be $+1$. The invertibility of 2 makes this a contradiction. Thus no linked folding arising from adjacent roots occurs here.

Let $R_0 = (R^\theta)^\circ$. The identity component of a pinned fixed group is reductive, with maximal torus $(T^\theta)^\circ$, and its rank is 2 or 4, respectively. The order of the component group of the full fixed group is prime to p: lift the action to the product of the simply connected cover of the derived group and the central torus. The pinned fixed group of the derived group

is connected, the fixed-torus component group is killed by e, and the relevant central-isogeny degrees for the root subsystems are prime to p. Consequently $\Gamma_u \in R_0$. In characteristic zero, the algebraic group generated by a unipotent element is connected. In characteristic p, its image in the finite component group has p-power order and hence must be trivial.

Now (5) implies

$$\dim Z_{R_0}(\Gamma_u) = \operatorname{rank} R_0. \tag{14}$$

Thus Γ_u is regular unipotent in R_0. Conjugate it within R_0 into the unipotent radical of the fixed positive Borel. For an orthogonal orbit $\mathcal{A}$, the folded simple-root subgroup is parametrized by $\prod_{\alpha\in\mathcal{A}} x_\alpha(a)$. These root groups commute, so its image in $U_R/[U_R, U_R]$ has the nonzero diagonal coordinate in every direction of the orbit. The regular-unipotent criterion requires every folded simple coordinate to be nonzero. Hence every simple-root coordinate in the original group R is nonzero as well. This is exactly the ordinary regular-unipotent criterion in R. A cyclic permutation of isomorphic factors gives diagonal principal unipotents in all factors, and torus factors impose no additional condition.

Therefore Γ_u is regular in R, and

$$\dim Z_S(\Gamma) = \dim Z_R(\Gamma_u) = \operatorname{rank} R = \operatorname{rank} S. \tag{15}$$

This proves the first assertion of Lemma 9.1. The argument neither presupposes ordinary regularity of Γ nor restricts the spectrum of Γ_s, and it does not require the residual element to be semisimple.

9.8 The full centralizer and the injection of its central cocycles

Put $C = Z_S(\Gamma)$. In the allowed characteristics, the structure of a regular-unipotent centralizer gives

$$C = Z(R) \times U, \tag{16}$$

where U is connected and unipotent. Here $Z(R)$ is the full smooth diagonalizable center, including any finite components; it cannot be replaced by $Z(R)^\circ$. For the tame cyclic action, $H^1(C_e, U(k))$ is trivial, so $H^1(C_e, C(k))$ reduces to $H^1(C_e, Z(R)(k))$.

The character lattice of $T/Z(R)$ is the root lattice of R, and θ permutes its simple-root basis. Thus $(T/Z(R))^\theta$ is a connected torus. Taking invariants under a finite tame group is exact on vector spaces, so the differential of $T^\theta \longrightarrow (T/Z(R))^\theta$ is surjective. Its image is a closed subgroup of full dimension in a connected target and therefore is surjective on k-points. The cohomology exact sequence yields

$$H^1(C_e, C(k)) = H^1(C_e, Z(R)(k)) \hookrightarrow H^1(C_e, T(k)). \tag{17}$$

Since S is simply connected, $X_*(T) = Q^\vee$, the coroot lattice. The calculation of tame cyclic torus cohomology over an algebraically closed field gives

$$H^1(C_e, T(k)) \simeq (Q^\vee)^w / NQ^\vee, \qquad N = 1 + w + \cdots + w^{e-1}. \tag{18}$$

9.9 The integral norm lattice in E_6 and exclusion of the nontrivial classes

For (7) or its inverse, set $b = \alpha_2^\vee + \alpha_3^\vee + \alpha_5^\vee$. A direct calculation in the coroot basis gives

$$(Q^\vee)^w = \mathbb{Z}b \oplus \mathbb{Z}\alpha_4^\vee, \qquad NQ^\vee = \mathbb{Z}b \oplus 3\mathbb{Z}\alpha_4^\vee. \tag{19}$$

To make the integral index explicit, the six columns of $N\alpha_i^\vee$ in the basis $(b, \alpha_4^\vee)$ are

$$(-2,-3),\ (1,0),\ (1,0),\ (0,3),\ (1,0),\ (-2,-3). \tag{20}$$

Hence the cohomology is $\mathbb{Z}/3$, and the two nontrivial classes are represented by $\alpha_4^\vee(\zeta)$ and $\alpha_4^\vee(\zeta^2)$. The six fixed roots are

$$\pm\alpha_4, \qquad \pm(\alpha_2+\alpha_3+\alpha_4+\alpha_5), \qquad \pm(\alpha_2+\alpha_3+2\alpha_4+\alpha_5). \tag{21}$$

For either nontrivial representative, the phase multiplicities on these six roots are $(0,3,3)$: the phases $1,\zeta,\zeta^2$ occur $0,3,3$ times, respectively. For as, the 22 nonfixed three-cycles still contribute 22 fixed directions, the Cartan contributes 2, and the six fixed roots no longer contribute any fixed directions. Therefore

$$\dim Z_S(as) = 24 \neq 30 = \dim Z_S(s). \tag{22}$$

9.10 The integral norm lattice in E_7 and exclusion of the nontrivial classes

For (11), an integral basis of the invariant coroot lattice is

$$\begin{aligned} b_1 &= \alpha_1^\vee + \alpha_6^\vee, & b_2 &= \alpha_3^\vee + \alpha_5^\vee, \\ b_3 &= \alpha_2^\vee, & b_4 &= \alpha_4^\vee. \end{aligned} \tag{23}$$

The norms of the seven simple coroots, in their order, are

$$b_1,\ 2b_3,\ b_2,\ 2b_4,\ b_2,\ b_1,\ -2b_1-3b_2-2b_3-4b_4. \tag{24}$$

It follows that

$$(Q^\vee)^w/NQ^\vee \simeq (\mathbb{Z}/2)^2. \tag{25}$$

The three nontrivial classes are represented by $\alpha_2^\vee(-1)$, $\alpha_4^\vee(-1)$, and their product. On the twenty-four roots of the fixed D_4, the pairing

$$\big(\langle\alpha,\alpha_2^\vee\rangle, \langle\alpha,\alpha_4^\vee\rangle\big) \bmod 2$$

takes precisely the values $(1,0),(0,1),(1,1)$, each eight times. This follows directly from the D_4 simple system in (13). Thus every nontrivial representative has eight positive and sixteen negative phases on the fixed roots. Each of the 51 nonfixed pairs retains one fixed direction, so

$$\dim Z_S(as) = 4+51+8 = 63 \neq 79 = \dim Z_S(s). \tag{26}$$

9.11 Completion of the centralizer-conjugacy assertion in the finite lemma

Let s' be the competing lift in Lemma 9.1, and put $d = s's^{-1}$. The two identical almost-commuting relations imply $d \in C$. Since s' is conjugate to the actual s, their actual central powers agree: $s'^e = s^e$. Consequently

$$d\,\theta(d)\cdots\theta^{e-1}(d) = 1, \tag{27}$$

and d is a genuine cyclic cocycle in the full centralizer. First remove the unipotent part by conjugation in C, and then use torus conjugation to reduce its image to one of the representatives in (19) or (25). If its class were nontrivial, (22) or (26) would give centralizer dimension 24 or 63 for s', contradicting its conjugacy to the actual s. Its torus class is therefore trivial. By (17), its class in the full centralizer is trivial as well. Being a coboundary means precisely that $s' = csc^{-1}$ for some $c \in C(k)$. This completes the proof of Lemma 9.1.

9.12 Checking every geometric condition of tame integral descent

Apply the uniform tame integral descent lemma of Section 8 to the parameters of Subsection 9.2. We give the checks and the actual descent relation here to show that the finite lemma handles the regular partial fiber in the original group, rather than only a calculation over the residue field.

Let $F = L(u)$, where $u^e = t$ and $\sigma(u) = \zeta u$, and set

$$A = \mathrm{Ad}(q(u)), \qquad \delta = \mathrm{Ad}(s)\sigma. \tag{28}$$

For E_7, the adjoint cocharacter defines A, and the resulting automorphism of the simply connected group S_F is well defined. The table of actual lifts gives

$$\delta^e = 1, \qquad \delta A = A\sigma, \qquad v_F(u^{\alpha(q)}a) = e\, v_L(a) + \alpha(q). \tag{29}$$

Thus, under the identification by A, the tame fixed model

$$\left(\mathrm{Res}_{\mathcal{O}_F/\mathcal{O}}\, S_{\mathcal{O}_F}\right)^{\delta}$$

has root-subgroup bounds $v_L(a) \geq \lceil -\alpha(x) \rceil$, exactly those of the connected parahoric at x. Smoothness of the fixed model follows by averaging infinitesimal lifting obstructions with $1/e$. Its reductive quotient is $Z_S(s)$ in (4), and the kernel of reduction is a connected unipotent jet group. Since semisimple centralizers in the simply connected group are connected, this model is not a full facet normalizer with additional components. In H^+, the corresponding reductive quotient is $(Z_S(s) \times \mathbb{G}_m)/Z(S)$. The full facet stabilizer modulo its connected parahoric is $\mathbb{Z}$, measured by $v(\chi)$; only its diagram action factors through $\mathbb{Z}/e$.

Suppose the fixed element is $\gamma \in P^+\dot{\tau}^r$, with $e \nmid r$. Write $\chi(\gamma) = t^r a$, choose $\eta \in \mathcal{O}^*$ satisfying $\eta^e = a$, and express it over F as $\gamma = [h, u^r\eta]$. The sign in the central quotient (2) gives

$$\sigma(h) = z^{-r}h, \qquad X = A(h) \in S(\mathcal{O}_F), \qquad \delta(X) = z^{-r}X. \tag{30}$$

Here integrality of X follows because γ fixes x: after applying A, it fixes the hyperspecial origin over F. Only after reduction modulo u do we obtain $s\overline{X}s^{-1} = z^{-r}\overline{X}$. The semilinear relation (30) over the integral ring cannot be replaced by an ordinary almost-commuting relation.

The element $\overline{X}$ lies in the actual disconnected normalizer coset checked in Subsection 9.2: triality of D_4 or diagram reversal of E_6, with no fixed directions on the additional center. Regularity in the original residual coset for H^+ is therefore equivalent to $\dim Z_S(s, \overline{X}) = 2$ or 4. The scalar center adds one dimension in each case, and the finite central quotient does not change dimensions. Lemma 9.1 therefore makes each such $\overline{X}$ an ordinary regular element.

Take two regular flag points. First use the identity-component centralizer torus to move their full $\pi_1(H^+) = \mathbb{Z}$ grades to zero. Surjectivity of the torus Kottwitz map from Section 3 ensures that this retains all central labels. The e-th root of their scalar unit can be absorbed into the parahoric, so choose representatives $h_1, h_2 \in S(L)$. Set

$$X_i = A(h_i)^{-1} X A(h_i).$$

They are integral, conjugate over F, satisfy the same relation (30), and have ordinary regular reductions.

The integral Steinberg quotient of a split simply connected group is smooth at these points. Fundamental characters give integral quotient coordinates, whose differentials have full rank at regular residual points. A regular fiber is a single geometric conjugacy orbit, and

the kernel of the quotient differential is the tangent space to that orbit. Consequently the relative action morphism

$$S \times S^{\mathrm{reg}} \longrightarrow S^{\mathrm{reg}} \times_{S//S} S^{\mathrm{reg}} \tag{31}$$

is smooth. Equality of quotient invariants on the generic fiber is equality of the integral invariants. The transporter thus has a k-point and, by complete smooth lifting, an element $a \in S(\mathcal{O}_F)$ with $aX_1a^{-1} = X_2$. The full group scheme $J = Z_{S_{\mathcal{O}_F}}(X_1)$ is smooth and is δ-stable by the central-multiple relation (30).

The error $d = a^{-1}\delta(a)$ is a cyclic cocycle in $J(\mathcal{O}_F)$. Modulo u, it satisfies

$$\overline{d} = s's^{-1}, \qquad s' = \overline{a}^{-1}s\overline{a}. \tag{32}$$

The actual s' is conjugate to the actual s and satisfies the same relation with $\overline{X}_1$. The second assertion of Lemma 9.1 supplies a coboundary in the full $J(k)$, killing the error modulo u. Smoothness permits the correcting element to be lifted integrally. Every subsequent congruence layer is a Lie-algebra vector group. Its cyclic action includes the factor ζ^{ℓ} contributed by the uniformizer at that layer, and its H^1 vanishes by averaging with $1/e$. Successive corrections and completeness produce a δ-fixed a_∞ still satisfying $a_\infty X_1 a_\infty^{-1} = X_2$.

By (29), $a_\infty = A(a_L)$, and the root bounds place a_L in the connected parahoric of $S(L)$. Thus $h_2 a_L h_1^{-1}$ centralizes the original γ and transports the two original flag points. The generic regular semisimple centralizer in H^+ is a connected torus, so this transporter belongs to the identity-component centralizer. Restoring the initial torus grade translations keeps it in that same identity torus. When $e \mid r$, a scalar central element of grade e moves the fixed element to grade zero, after which the ordinary hyperspecial case applies. Thus every basic grade of H^+ has been covered.

These checks apply identically to $\mathcal{O} = k[[t]]$ and $\mathcal{O} = W(k)$: e is invertible, strict Henselianity supplies the required roots of unity and e-th roots of units, and the integral Steinberg quotient and smooth lifting apply in both settings. We have not compared numbers of affine Springer components over F and over L.

9.13 Transfer to adjoint E_6, E_7 and arbitrary central forms

The kernel of $H^+ \longrightarrow S/Z(S)$ is the split scalar torus $\mathbb{G}_m$. Hilbert 90 lifts the fixed element and the variable flag points. The injection of the kernel's cocharacter lattice into $\pi_1(H^+)$ ensures that any extra central factor after lifting has valuation zero and can be absorbed into the parahoric. Regularity is preserved under central quotients. The local transitivity of Subsection 9.12 therefore descends to the adjoint group, handling the two nonzero classes in E_6 and the unique nonzero class in E_7. The zero classes were handled in Subsection 9.1.

For any split reductive central form and any actual Newton Levi subgroup M, apply the universal central-form transfer proved in Section 3. Specifically, take a split central-torus extension $H \longrightarrow M$ with simply connected H_{der}, and use the natural finite central isogeny

$$H \longrightarrow H_{\mathrm{ad}} \times (H/H_{\mathrm{der}}). \tag{33}$$

Its Kottwitz image is a graph inside the product of the adjoint fundamental group and the free abelianization lattice. The grade of a target identity-centralizer transporter is the difference of two grades allowed in the source, so it lifts to H. The inverse image of the target identity torus is a maximal torus of the source; consequently the lift still centralizes the fixed element. Finally descend to M through the split central kernel. This retains all central couplings among factors without requiring an arbitrary fixed element to lift beforehand to a specified finite simply connected cover.

9.14 Node-by-node classification of every proper Newton Levi

For a dominant Newton vector, the simple-root set of the standard Newton Levi is

$$J = \{\alpha : \langle \alpha, \nu_\gamma \rangle = 0\}.$$

Every proper J is contained in a root subset obtained by deleting a single node. The following complete single-node deletion table therefore also proves exhaustiveness for all proper Levi subgroups.

Ambient type	Deleted node	Semisimple type of the maximal proper Levi
G_2	Either node	A_1
F_4	1	C_3
F_4	2	$A_1 \times A_2$
F_4	3	$A_2 \times A_1$
F_4	4	B_3
E_6	1 or 6	D_5
E_6	2	A_5
E_6	3 or 5	$A_1 \times A_4$
E_6	4	$A_1 \times A_2 \times A_2$
E_7	1	D_6
E_7	2	A_6
E_7	3	$A_1 \times A_5$
E_7	4	$A_1 \times A_2 \times A_3$
E_7	5	$A_2 \times A_4$
E_7	6	$D_5 \times A_1$
E_7	7	E_6
E_8	1	D_7
E_8	2	A_7
E_8	3	$A_1 \times A_6$
E_8	4	$A_1 \times A_2 \times A_4$
E_8	5	$A_3 \times A_4$
E_8	6	$D_5 \times A_2$
E_8	7	$E_6 \times A_1$
E_8	8	E_7

For F_4, the numbering is $1 - 2 \Rightarrow 3 - 4$, with subdiagram $\{1, 2, 3\}$ of type B_3 and subdiagram $\{2, 3, 4\}$ of type C_3. Further deletion in type A produces only smaller type-A factors, and deletion in types B, C, D produces only smaller classical factors. Applying the preceding table recursively gives the following precise bounds on the possible factors.

Ambient type	Possible simple factors in a proper Levi
G_2	A_1
F_4	$A_1, A_2, B_2 = C_2, B_3, C_3$
E_6	A_1 through A_5, D_4, D_5
E_7	A_1 through A_6, D_4 through D_6, E_6
E_8	A_1 through A_7, D_4 through D_7, E_6, E_7

Here $D_2 = A_1 \times A_1$ and $D_3 = A_3$; tori and the actual central couplings are allowed throughout. In particular, a proper E_8 Levi has no E_8 factor, a proper E_7 Levi has no E_7

factor, and a proper E_6 Levi has no E_6 factor. Thus the exceptional induction in this section is not circular.

9.15 The actual GSp_6 Levi in F_4

The group F_4 is both simply connected and adjoint, and its character lattice is the root lattice. For a standard Levi, quotienting by the simple roots in J quotients this free lattice by part of a basis. Hence the Levi center is connected. Moreover, the derived subgroup of a Levi in a simply connected group is simply connected.

For the C_3 Levi obtained by deleting node 1, the derived group is Sp_6 and the connected center has rank one. Its $\mu_2 = Z(\mathrm{Sp}_6)$ lies in the actual central torus, so the finite central quotient supplied by multiplication is

$$M_{C_3} \simeq (\mathrm{Sp}_6 \times \mathbb{G}_m)/\mu_2 \simeq \mathrm{GSp}_6. \tag{34}$$

Likewise the maximal B_3 Levi is GSpin_7. The C_3 Newton Levi of F_4 therefore has basic classes with odd similitude valuation; the theorem for Sp_6 alone does not handle these classes. The local theorem for GSp_{2a} in Section 4 includes odd multiplier grades and every very special chain, and hence provides the actual local input for (34). The B_3 case and its central forms are handled by the orthogonal results of Section 5 and the transfer of Section 3.

9.16 Every Newton point in exceptional types

All basic adjoint classes for classical factors of types A, B, C, D were covered in the preceding sections, including the vector class and both spinorial classes in type D, and the odd similitude class in type C. All basic adjoint classes in E_6, E_7 are covered by Subsections 9.1–9.13. The full basic cases for G_2, F_4, E_8 are covered by Subsection 9.1. Subsection 9.14 exhausts all proper Newton Levi subgroups, and Subsection 9.13 transfers the local property to each actual central form. Thus every actual Newton Levi of exceptional type, every basic Kottwitz class, and every very special regular endpoint satisfy local transitivity under the identity-component centralizer torus.

By the exact component reduction of He in Section 2, the set $\mathcal{R}$ of actual top-dimensional paths for $w = w_0 t^\mu$ satisfies

$$C_\mu(\gamma) = \sum_{a \in \mathcal{R}} n_{a,\gamma}, \qquad |\mathcal{R}| = \dim V_\mu(\lambda_b). \tag{35}$$

Every actual top-dimensional endpoint is very special. Local transitivity and the description of regular partial-fiber components give $n_{a,\gamma} = 1$. Consequently

$$\#\big(Z_G(\gamma)(L)\backslash \Sigma^{\mathrm{top}} Y_\mu(\gamma)\big) = \dim V_\mu(\lambda_b). \tag{36}$$

Here λ_b is always the best integral approximation associated with the relevant $[b]$, retaining its full Kottwitz data, rather than the rational Newton vector. The conclusion includes every exceptional simple type, all its split reductive central forms, and every Newton point.

10 All Root Types, Actual Newton Levis, and Completion

10.1 The local property for every simple adjoint type

The preceding results give the exhaustive table below.

Simple adjoint type	Basic grades and local proof
A_r	The general linear cyclic-lattice theorem in Section 4, descended through the split scalar kernel of $GL_{r+1} \to PGL_{r+1}$; every rotation class is included.
B_r	The adjoint group is SO_{2r+1}; Section 5 includes the ordinary and nonzero vector classes.
C_r	The even and odd multiplier classes of GSp_{2r} in Section 4, descended through its split scalar kernel to PSp_{2r}.
D_r	The ordinary and vector cases of Section 5, the finite lemmas of Sections 6 and 7, and the descent of Section 8, including both half-spin and primitive order-four classes.
G_2, F_4	Trivial adjoint fundamental group; the full basic case is ordinary hyperspecial transitivity.
E_6, E_7	The finite lemmas of Section 9 and tame integral descent, including both nonzero E_6 classes and the nonzero E_7 class.
E_8	Trivial adjoint fundamental group; the full basic case is ordinary hyperspecial transitivity.

The small-rank identifications $A_1 = B_1 = C_1$, $D_2 = A_1^2$, and $D_3 = A_3$ are already included. A root-free group is a split torus; its flag points form the cocharacter lattice and are transitive under torus L-points.

Every local transporter lies in the identity centralizer of the regular semisimple fixed element. The simply connected derived envelopes have connected generic torus centralizers; the ordinary SO proof constructs its generic identity-torus model directly; and Section 3 transfers only through those tori. Connectedness of the full centralizer in an arbitrary adjoint group has not been asserted.

10.2 Every actual Newton Levi

Let G now be an arbitrary split connected reductive group and let γ be any regular semisimple element. For each actual top path $\mathfrak{p}$, He's Theorem 1.3 moves the fixed element into a basic problem in the actual split Newton Levi $M_{\mathfrak{p}}$. Its adjoint group is a product of the simple types in the table. Log-volumes add over root factors, so a maximum must be attained in every factor; coset regularity and identity-centralizer actions factor as well. The table and split-torus translation give the property on the adjoint product times its abelianization. The central extension and graph isogeny of Section 3 transfer it to $M_{\mathfrak{p}}$ itself, retaining its actual central coupling. Consequently

$$\mathsf{L}(M_{\mathfrak{p}}), \qquad n_{\mathfrak{p},\gamma} = 1$$

for every such path. There is no bound on Newton slopes or central grades: full integer Kottwitz grades and finite diagram actions are recorded separately in the cyclic envelopes.

10.3 The component formula

Substitution in the exact formula of Section 2 gives

$$\begin{aligned}\#\left(Z_G(\gamma)(L)\backslash\Sigma^{\mathrm{top}}Y_\mu(\gamma)\right) &= \sum_{\mathfrak{p}\in\mathcal{R}} n_{\mathfrak{p},\gamma}\\ &= \sum_{\mathfrak{p}\in\mathcal{R}} 1\\ &= |\mathcal{R}|\\ &= \dim V_\mu(\lambda_b).\end{aligned}$$

This completes the assembled argument for Theorem 1.1. It imposes neither integral Newton nor regular μ, only nonemptiness, regular semisimplicity of γ, and the field, splitness, and characteristic assumptions stated at the beginning.

10.4 All grades of the orthogonal central forms

For a concrete check on PSO, GSO, and half-spin forms, write $N = 2n \geq 4$. The D_n coroot and adjoint cocharacter lattices are

$$Q = \{a \in \mathbb{Z}^n : \textstyle\sum a_i \text{ even}\}, \qquad P = \mathbb{Z}^n \cup (\mathbb{Z} + \frac{1}{2})^n.$$

Thus

$$\pi_1(PSO_{2n}) = P/Q = \begin{cases}\mathbb{Z}/4, & n \text{ odd},\\ (\mathbb{Z}/2)^2, & n \text{ even}.\end{cases}$$

The vector class is represented by e_1; the two spinorial classes by the spinor-end fundamental coweights. In odd rank they are a primitive order-four generator and its inverse; in even rank they are the two nonvector order-two classes. Sections 5–8 treat them separately.

For GSO_{2n}, split torus cocharacters are $(a_1, \ldots, a_n; c)$, with natural weights $a_i, c - a_i$ and multiplier valuation c. Coroots have $c = 0$ and even coordinate sum, so

$$\pi_1(GSO_{2n}) = \mathbb{Z} \oplus \mathbb{Z}/2, \qquad \kappa = (c, \textstyle\sum_i a_i \bmod 2).$$

Its adjoint image is

$$[a - \tfrac{c}{2}(1, \ldots, 1)] \in P/Q.$$

The scalar central cocharacter has class $(2, n \bmod 2)$. Odd multiplier valuation therefore gives a genuine spinorial adjoint class; no unavailable L-square root is used to reduce it to SO.

In odd dimension there is an actual isomorphism

$$GSO_{2n+1} \simeq SO_{2n+1} \times \mathbb{G}_m.$$

For multiplier λ, the scalar $c = \det(g)/\lambda^n$ satisfies $c^2 = \lambda$ and $c^{-1}g \in SO$, without arbitrary root extraction. Odd-dimensional PSO equals SO. The two half-spin quotients

$$HSpin^{\pm}_{4m} = \mathrm{Spin}_{4m}/C_{\pm}$$

have fundamental group $\mathbb{Z}/2$, with images the two spinor subgroups in the adjoint group, and are both included. In odd rank D_{2m+1}, the half-spin representations are faithful on the order-four center; there is no additional half-spin order-two quotient. The unique order-two central quotient is SO_{4m+2}.

The low-dimensional cases SO_1, SO_2, GSO_1, and GSO_2 are trivial or split tori. Under the projective-image convention $PSO_2 = SO_2/\{\pm 1\}$ is a split torus; under quotient by its full center it is trivial. Also

$$PSO_4 = PGL_2^2, \qquad PSO_6 = PGL_4.$$

The four-dimensional half-spin forms are $PGL_2 \times SL_2$ and $SL_2 \times PGL_2$, and the eight-dimensional ones are isomorphic to SO_8 by triality. These identifications agree with the general argument.

11 Inputs, Finite Checks, and References

11.1 Dependency structure

All new nonintegral-Newton steps have been expanded in the text. The standard finite-dimensional and local structural results listed in Section 1, and He's geometric reduction, are inputs; their entire background theories are not reproved here. The dependency structure is

$$\text{actual top-path component formula} \impliedby \text{local property in every actual Newton Levi,}$$

$$\impliedby \text{simple adjoint local properties} + \text{central extensions and Kottwitz graph transfer.}$$

The new spinorial and exceptional local properties require

$$\text{finite almost-commuting lemma} + \text{integral Steinberg transporter}$$
$$+\text{full residue-cocycle correction} + \text{tame congruence averaging.}$$

Ordinary SO uses the Lie quotient with its Pfaffian; types A, C use cyclic lattices and integral norm correction in the original representation. All links are necessary. A Weyl-class or Levi inventory alone does not prove geometric transitivity; generic conjugacy alone does not give integral conjugacy; and retaining only the identity residue centralizer can omit finite central obstructions.

11.2 Reproducible finite root data

The following data were checked in the supporting exact computations. Their corresponding analytic proofs or standard finite classifications appear in the text.

1. All affine-A rotation endpoints, affine-C reversal endpoints, and ordinary/vector B, D log-volume maxima.
2. Even-D invariant and norm lattices and all invariant sign patterns. Checks in D_4 through D_{16} make every nonzero class reverse the nonzero half-spin character.
3. Odd-D norm lattices, fourth-root representatives, and square-spectrum dimensions, including D_5, D_7, D_9, D_{11}. The proof has no rank cutoff.
4. All 51 840 elements of $W(E_6)$ and its three order-three classes, of fixed ranks $0, 2, 4$ and sizes $80, 480, 240$.
5. The two E_7 orthogonal-triple involution classes, with $24, 8$ fixed roots and sizes $315, 3780$. The text also proves exhaustiveness using perpendicular systems D_6 and $D_4 + A_1$.
6. The E_6, E_7 invariant/norm lattices and every nonzero cocycle's fixed-root phases.
7. All 128 standard simple-root subsets in each principal E_7 central form, all 256 subsets in E_8, and the 57 factor/diagram-action pairs appearing in the inventory.

For example, the full E_6 Weyl check starts at the identity integral Cartan matrix and closes under its six simple reflections until no new matrix appears. It gives 51 840 matrices. Filtering nonidentity matrices whose cubes are the identity and closing representatives under

simple-reflection conjugation gives the three stated classes. Roots are generated by reflection closure from positive and negative simple roots; fixed-root counts and norm lattices use integer linear algebra. Matrix columns are roots, and their coefficient bounds are checked from the highest root, so the encoding neither samples nor uses floating-point approximations.

These calculations do not replace the finite regularity, smooth transporter, or descent arguments. This is a written mathematical argument, not a formal proof certificate.

11.3 Scope and manuscript status

The statement is restricted to the split groups, ground fields, and Weyl-prime characteristic hypothesis stated in Section 1, without further Newton or central-form restrictions. The standard inputs and all new local deductions are identified. Internal reviews and exact finite computations are not external peer review, publication recognition, or Lean formal certification.

11.4 Bibliographic sources

He theorem numbers throughout refer to the specified arXiv version. The Chen–Zhu count and very-special stabilizer result are invoked through He's Theorem 3.2.

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